DOORS & WINDOWS

DOOR

A **door** is a moveable barrier secured in a wall opening.

Functions:

1. They admit ventilation and light.
2. Controls the physical atmosphere within a space by enclosing it, excluding air drafts, so that interiors may be more effectively heated or cooled.
3. They act as a barrier to noise.
4. Used to screen areas of a building for aesthetic purposes, keeping formal and utility areas separate.

Location of door in a building

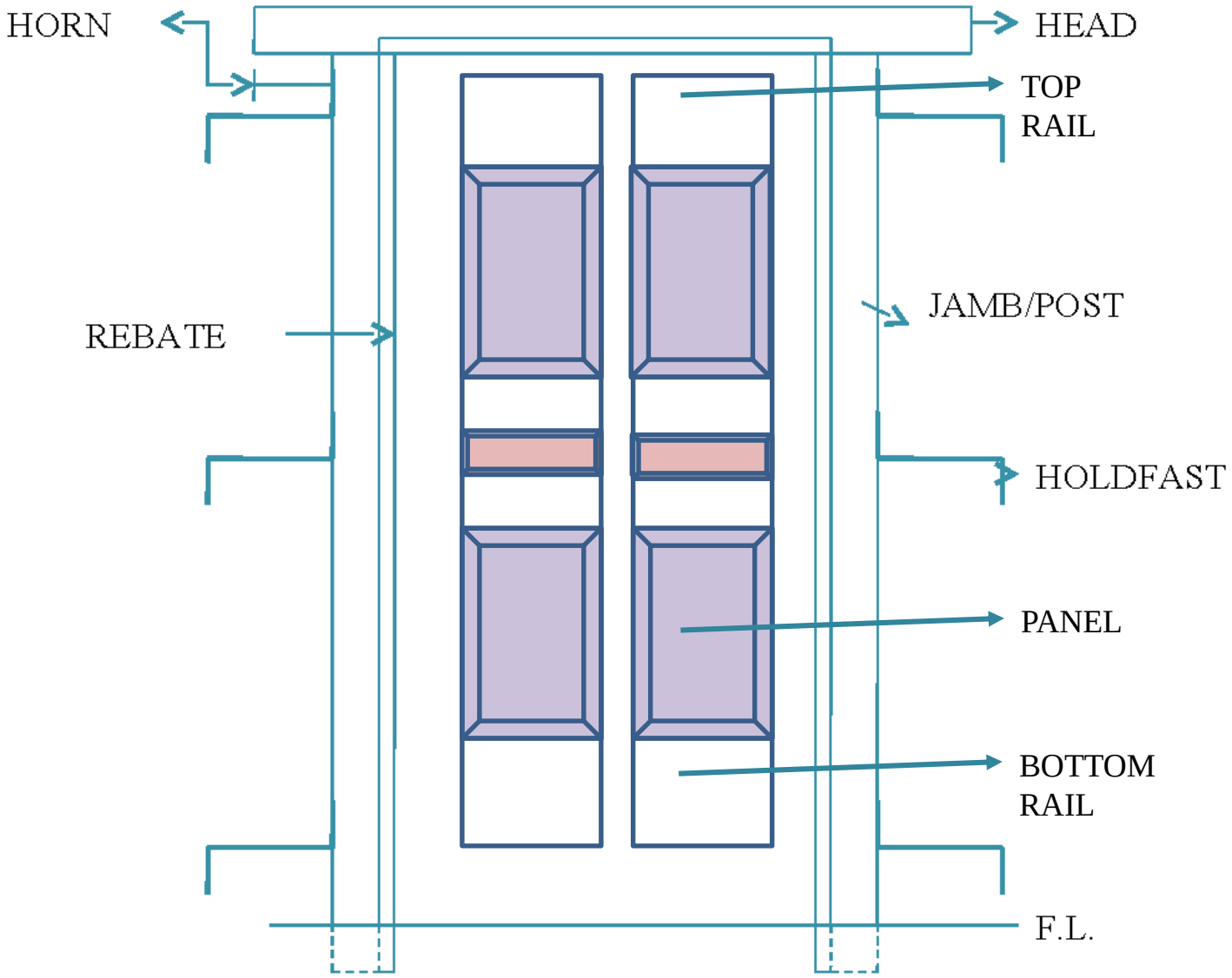
- The number should be kept as minimum.
- It should meet the functional requirement.
- It should preferably be located at the corner of the room, nearly 20 cm from corner.
- If in a room, more than 2 doors are there, they shall be located facing each other.

- Larger number of doors cause obstructions and consume more area in circulation.
- Door should not be located in center of length of wall.
- If there are 2 doors in room, door should preferably be located in opposite wall, facing each other so as to provide good ventilation and free air circulation.

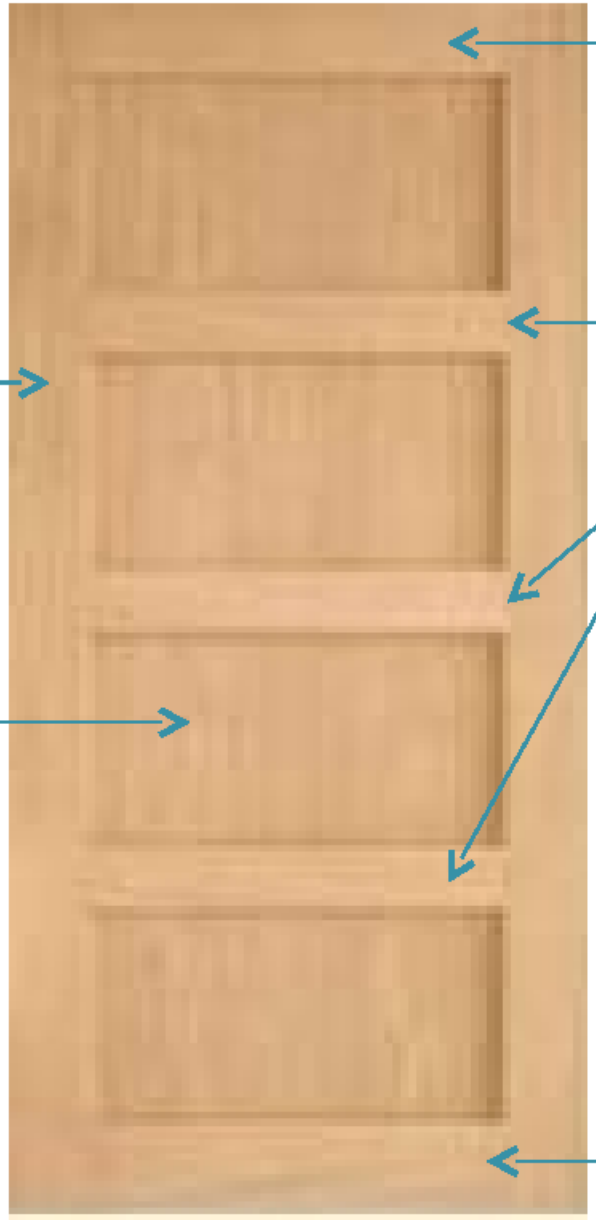
Components of a door:

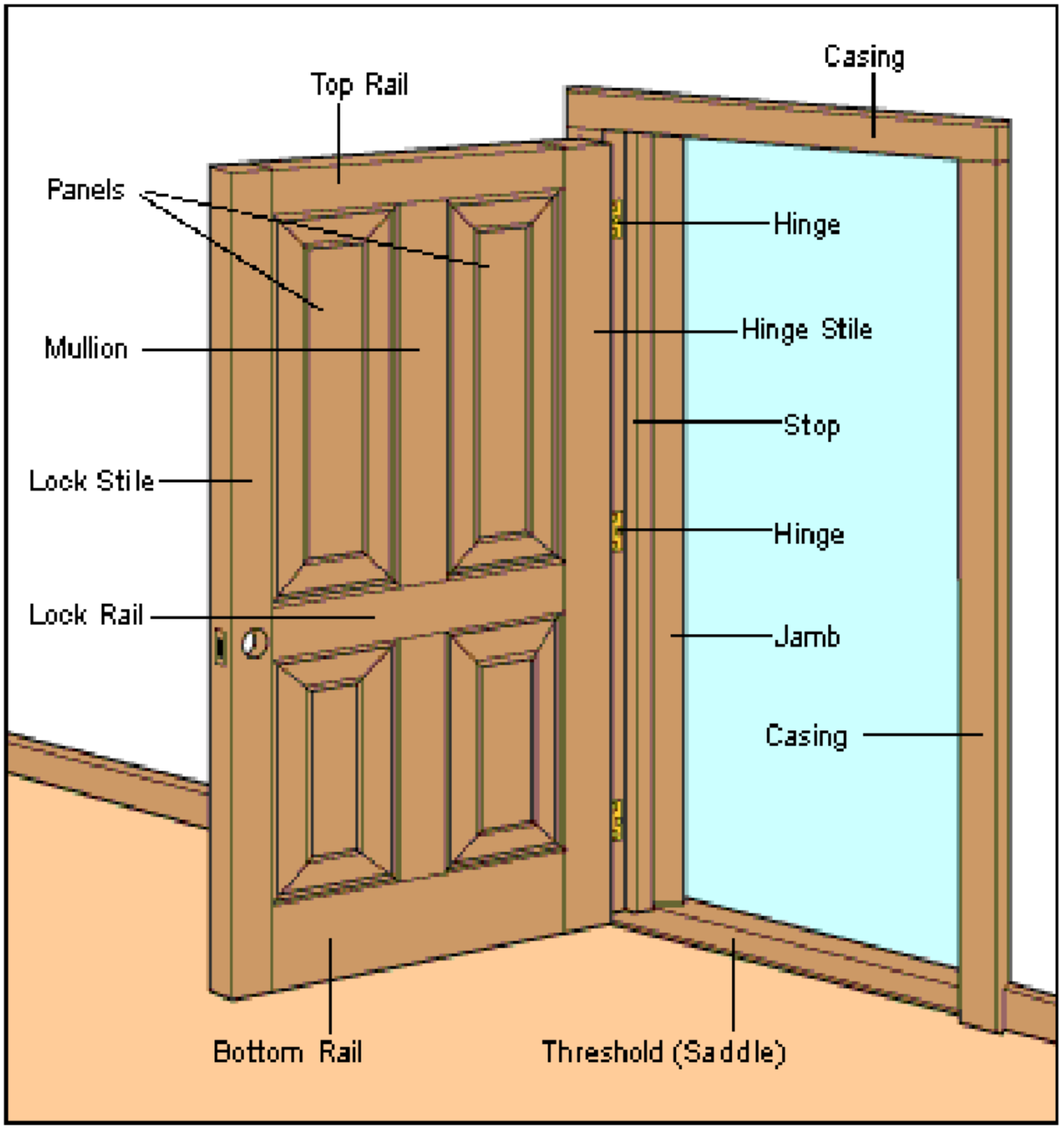
a) Door frame b) Door shutter

Door frame



Door shutter




Casing
Top Rail
Panels
Hinge
Hinge Stile
Mullion
Stop
Lock Stile
Hinge
Lock Rail
Jamb
Casing
Bottom Rail
Threshold (Saddle)

SIZES OF DOORS

The common width-height relations used:

- Width = 0.4 – 0.6 Height
- Height = (width +1.2)m

General sizes used:

a) Residential

External door – 1.0 x 2.0 to 1.1 x 2.0 m

Internal door - 0.9 x 2.0 to 1.0 x 2.0 m

Bath & WC – 0.7 X 2.0 to 0.8 x 2.0 m

Garages for cars – 2.25 x 2.25 m to 2.40 x 2.25 m

b) Public

1.2 x 2.0 m **or** 1.2 x 2.1 m **or** 1.2 x 2.25 m

Designation of doors

- The size of an opening is indicated by number of modules where each module is of 100 mm.

e.g. 10 DS 20 OR 10 DT 20

size of opening = 1000 x 2000 mm

D – Door opening

S – Single shutter

T – Double shutter

Recommended Dimensions for Doors

Sr. No	Designation	Size of opening(mm)	Size of Door frames(mm)	Size of Door Shutter(mm)
1	8DS20	800 X 2000	790 X 1990	700 X 1905
2	8DS21	800 X 2100	790 X 2090	700 X 2005
3	9DS20	900 X 2000	890 X 1990	800 X 1905
4	9DS21	900 X 2100	890 X 2090	800 X 2005
5	10DT20	1000 X 2100	990 X 1990	900 X 1905
6	10DT21	1000 X 2100	990 X 2090	900 X 2005
7	12DT20	1200 X 2000	1190 X 2090	1000 X 1905

DOOR FRAMES

Materials used for door frames

- Timber
- Steel
- Aluminium
- Concrete
- Stone

Timber door frame

General specifications:

- Timber is sawn in the direction of grains.
- All members of frames are of same species of timber and be straight without any warp.
- The frames are smooth, well planed surfaces except the surface touching wall lintel sill etc.
- The thickness of rebate is 15 mm and the width is equal to the thickness of shutter.
- Nominal size of door frame for single shutter is 75 X100 mm and for double shutter 75 X 125 mm.
- The back portion of door frame which in contact with walls, lintels sill etc. is painted with bitumen or any anti-termite chemical.
- To protect door frame during construction priming coat is done before fixing.
- A minimum of 3 holdfasts shall be fixed on each side, one at the centre and the other two at 300 mm from top and bottom of the frame.
- Holdfasts and other parts, which go into the masonry wall and thus not accessible for maintenance, shall be protected against moisture and decay, with a coating of coal tar or other suitable protective material.

Types of Doors

On the basis of working operations

- Hinged doors
  - Battened type
  - Framed and paneled
  - Glazed/Sash
  - Flushed
  - Louvered
  - Wire-guage
- Revolving doors
- Sliding doors
- Swing doors
- Collapsible doors
- Rolling shutter

Hinged doors

Most doors are hinged along one side to allow the door to pivot away from the doorway in one direction but not in the other. The axis of rotation is usually vertical.

The most common door type. It is a simple & rigid.

The panel swings, opens and closes, on hinges.

Hinged doors require a minimum amount of maintenance and cleaning, they are not expensive, and have an excellent insulating ability.

However, they take up precious room space to swing in.

Battened and ledged door

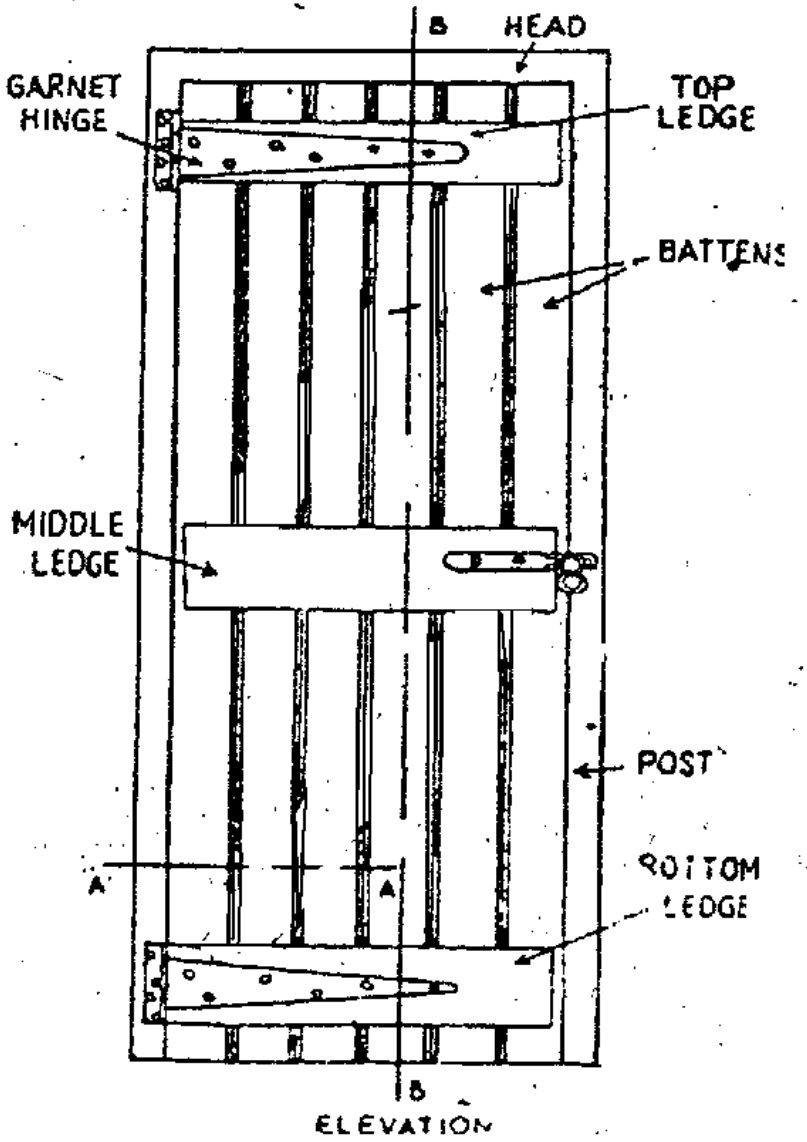
The door consists of vertical boards i.e. battens and three or four horizontal ledges. The vertical boards are tongue and grooved to stop draughts and the edges chamfered to relieve the plain appearance.

Battens : 100-150 mm wide and 20-30 mm thick

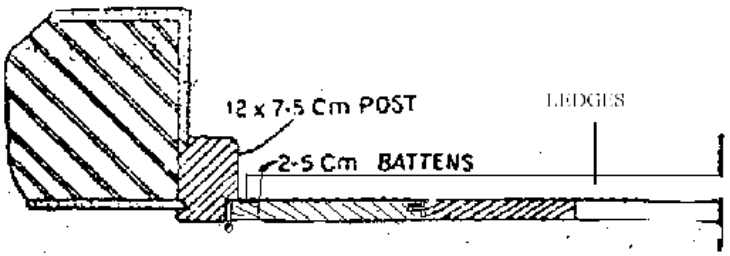
Ledges : 200 mm wide and 25 – 30 mm thick

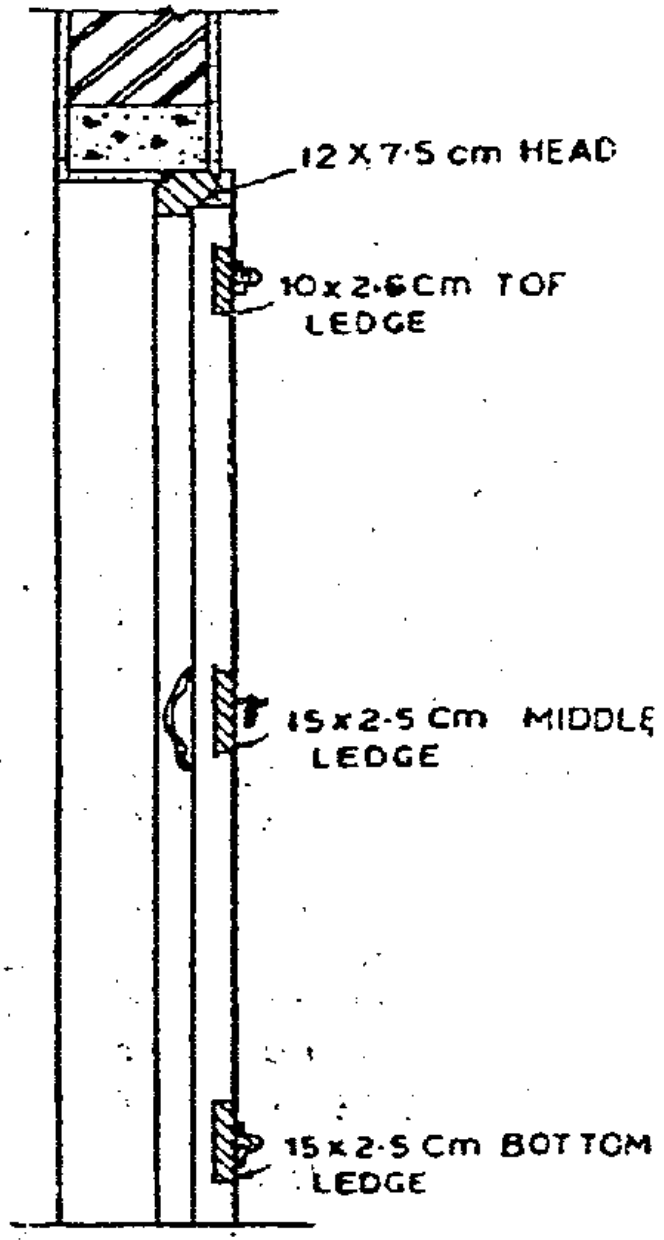
The door is hung to the frame by T-hinges of iron.

The door is commonly used for narrow openings for internal use where it is not subject to hard use, or where economy is of main consideration than the appearance..



ELEVATION





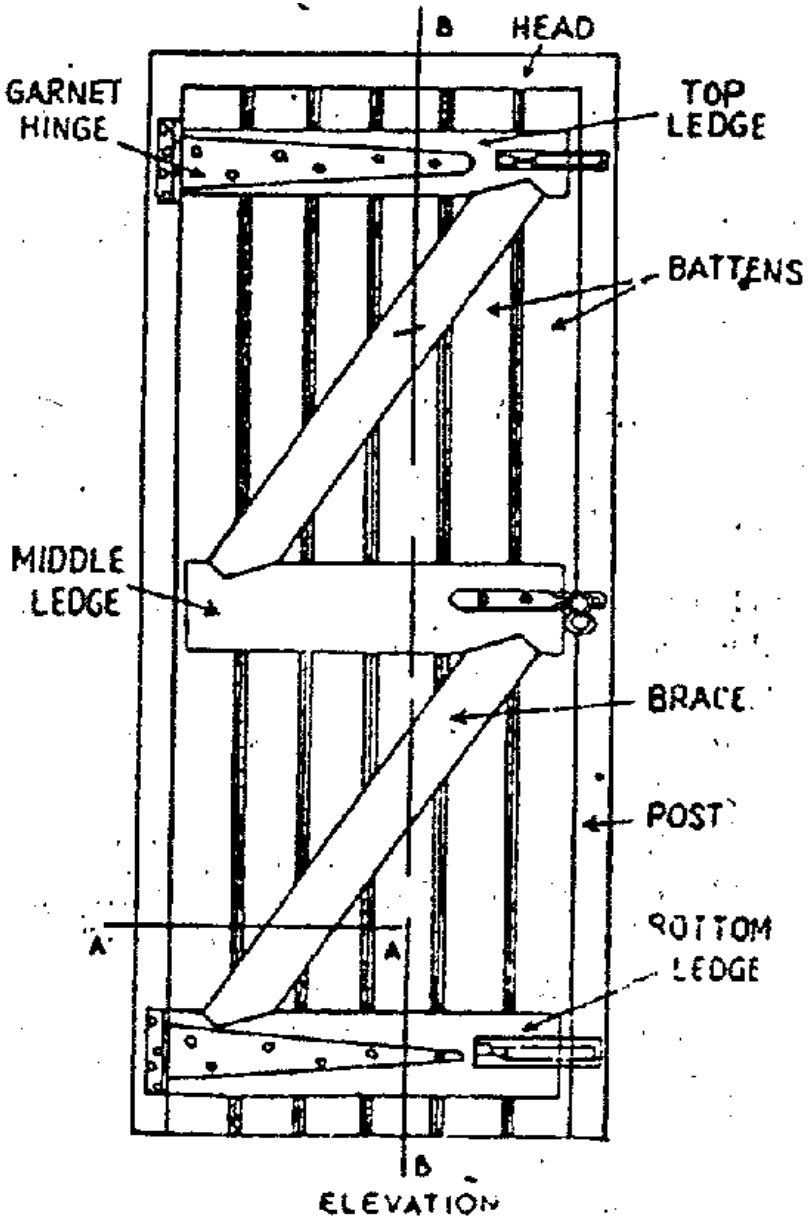
SECTION AT B. B

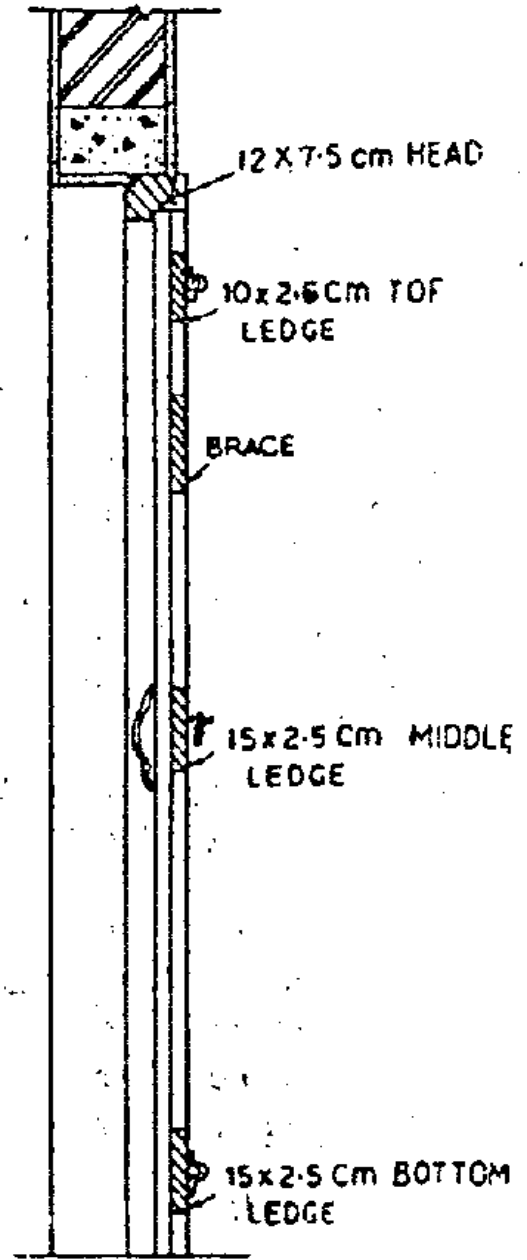
Battened, Ledged and Braced door

Battens : 100-150 mm wide & 20-30 mm thick

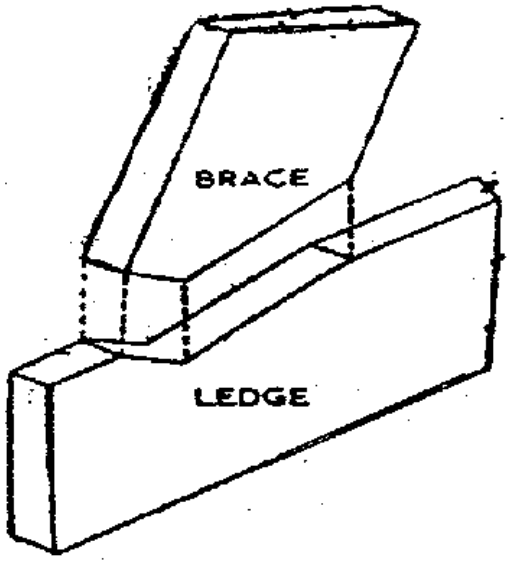
Ledges : 100-200 mm thick & 25-30 mm thick

Braces: 100-150 mm wide & 25-30 mm thick

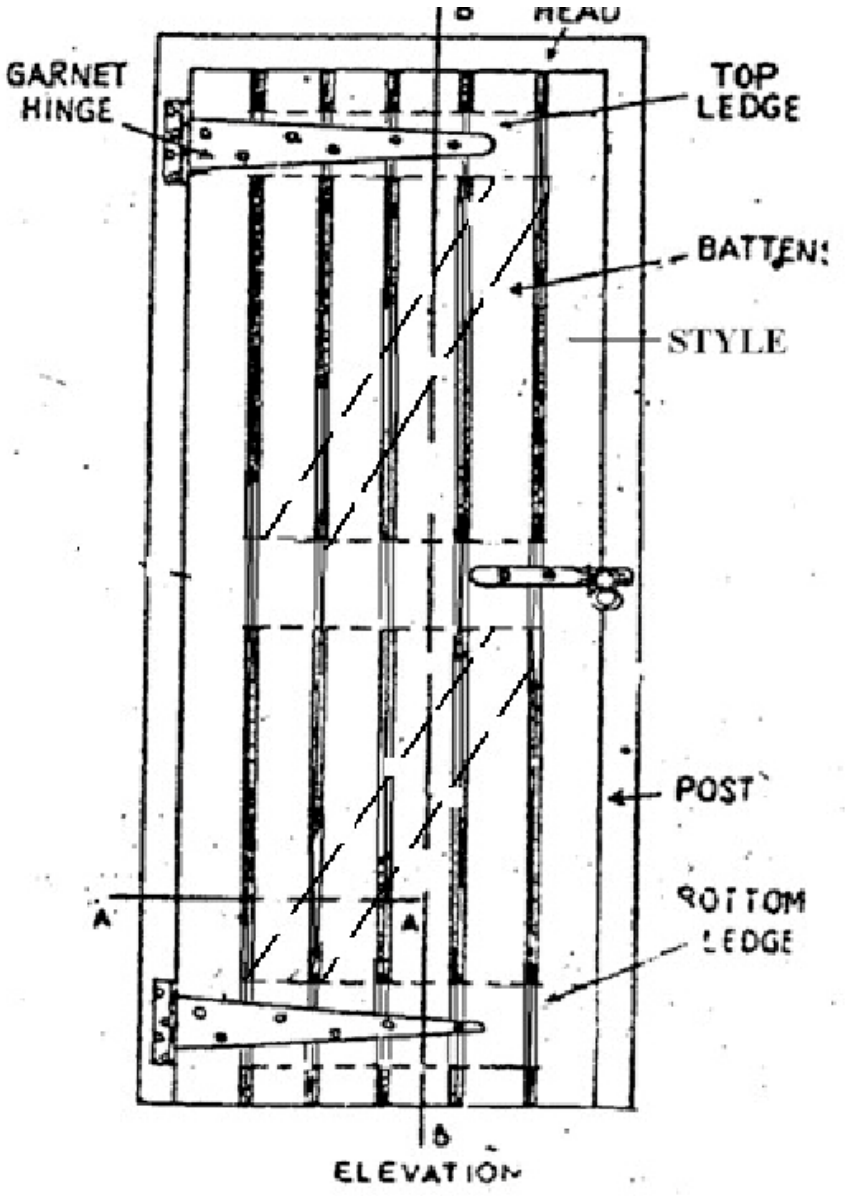




- Normally constructed using a Z-shaped frame with tongue-and-groove interlocking boards attached they can be quite heavy in weight but this can depend on the thickness of boards used.
- Due to their construction they are normally very strong and hardwearing and can also be planed and shaped to fit pretty much any door way.
- Such doors are used for wider openings.
- The braces incline down towards the hinged side.



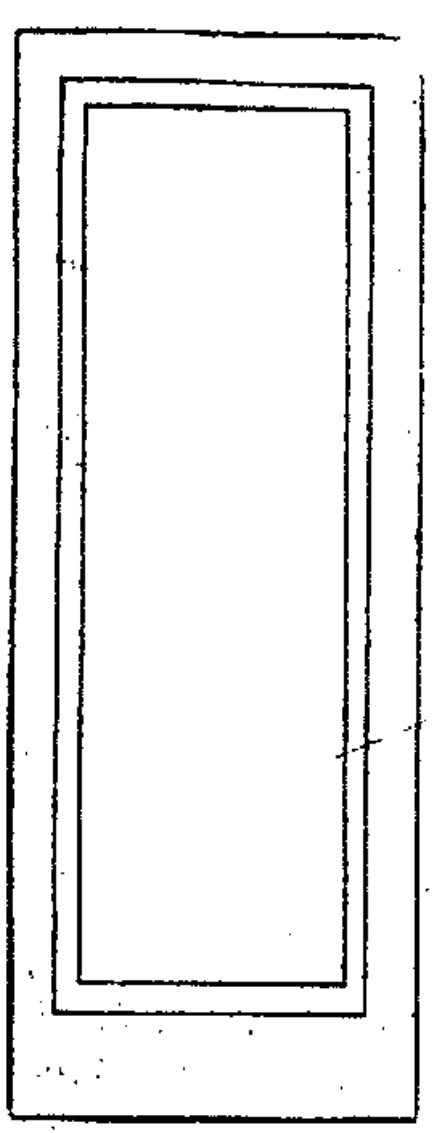
Battened, Ledged, Braced and Framed Door



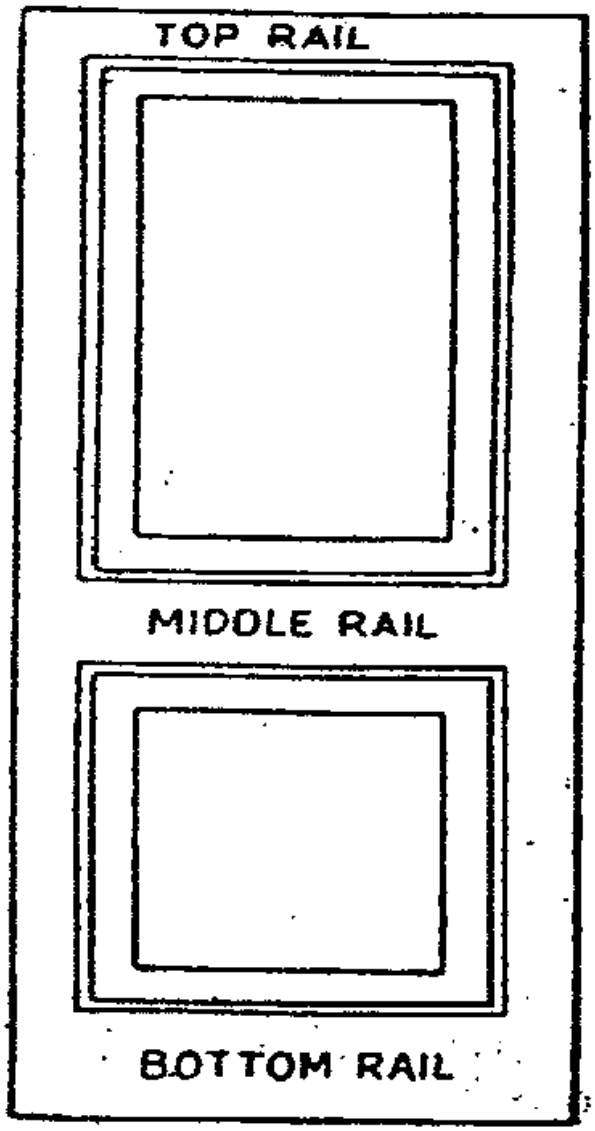
- Framework consist of vertical styles, three ledges and two inclined braces.
- The braces are normally housed into rails at about 40mm from the styles.
- The thickness of styles and top rail is same which is equal to the thickness of braces and battens.
- This type is suitable for external door and door subjected to rough handling.

Framed and Paneled door

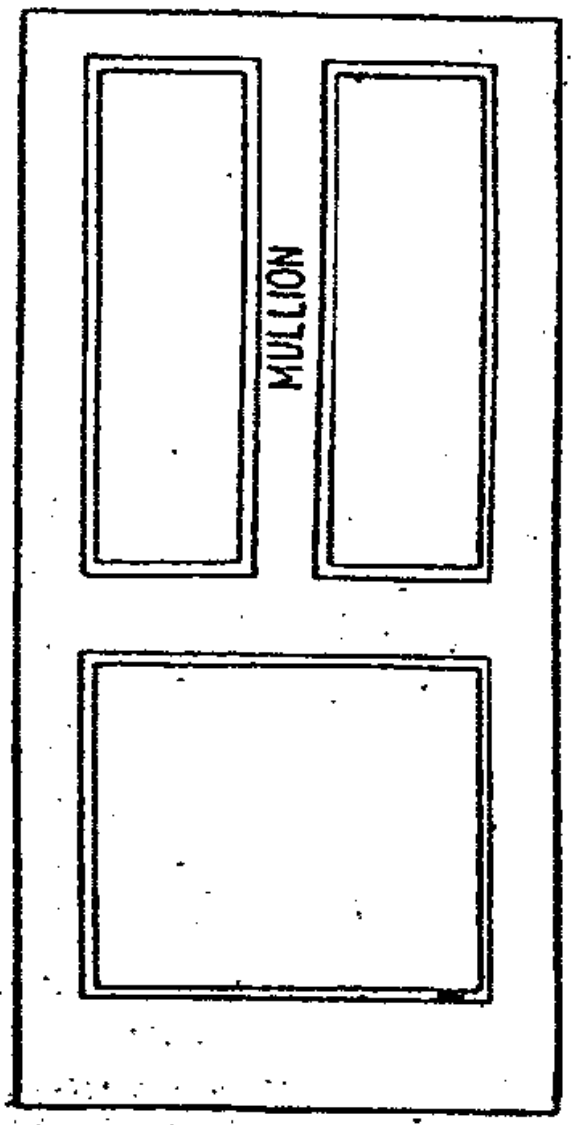
- These types of doors are widely used in all types of buildings since they are strong and give better appearance than battened doors.
- Panel doors consist of vertical members called stiles and horizontal members called rails.
- Stiles and rails form the framework into which panels are inserted.
- Panels may be solid wood, plywood, particleboard or louvered or have glass inserts.
- Additional vertical members called mullions are used to divide the door into any number of panels.



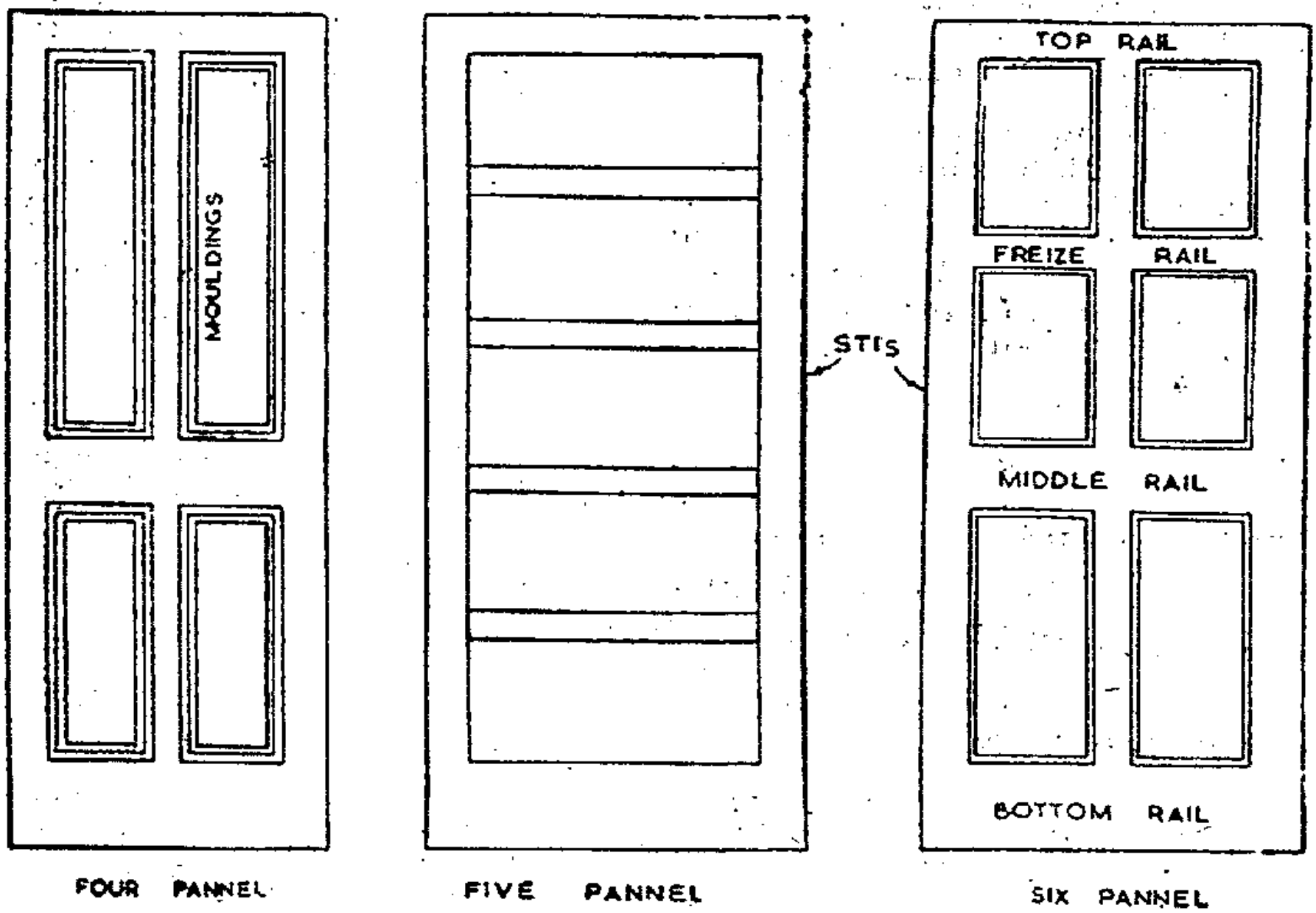
ONE PANNEL



TWO PANNEL



THREE PANNEL


MOULDINGS
STIS
TOP RAIL
FREIZE
RAIL
MIDDLE RAIL
BOTTOM RAIL
FOUR PANNEL
FIVE PANNEL
SIX PANNEL

- Panel Door - consists of wood panels held in place by wood stiles and rails panels and can be made from solid wood, plywood, metal or glass.
- They consist of two vertical stiles and horizontal rails enclosing the panels. The panels could be made of plywood, solid timber or glass.
- The more expensive type of glass panel door is fitted to the frame first and then the glass panels are added individually, this makes the door lighter to handle when hanging and cheaper to replace if the glass gets broken.
- Depending on the location and usage of the door, putty and beading may be used, but if the door has a wood finish, beading alone will suffice. Silicone sealants are also available in a variety of finishes.
- The lock rail is so placed that its centre is at a height of 800-900 mm from the bottom of the shutter.
- Incase of timber panels, the minimum size of the panel shall be 150 x 15 mm, while max. are shall be 0.5 sq. m.
- Incase of plywood, there is no restriction on minimum and max. size. However, the thickness shall be at least 12 mm if single paneled and 10 mm for more panels.

Glazed or sash door

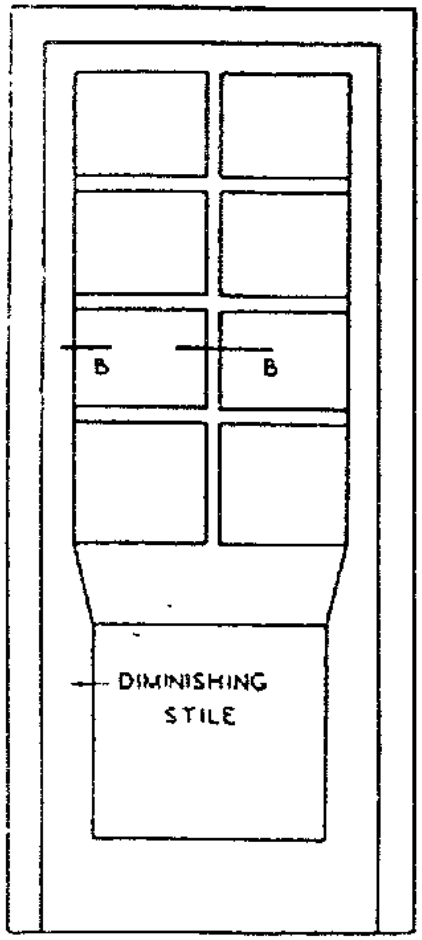
- This type is used in residential and public buildings.
- They supplement the natural lighting provided by windows or to make the interior of the room visible from adjoining rooms.
- They can be made fully glazed or partly glazed.
- Fully glazed doors are recommended where sufficient light is required through the door openings like in shopping malls, entrance halls etc.
- Incase of partly glazed, the bottom 1/3 rd part is usually paneled and upper 2/3 part is glazed.



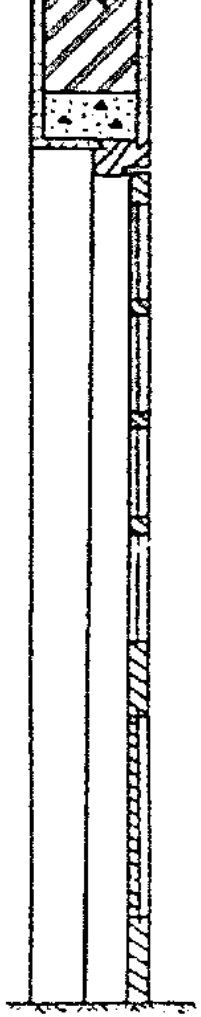
Fully Glazed Door



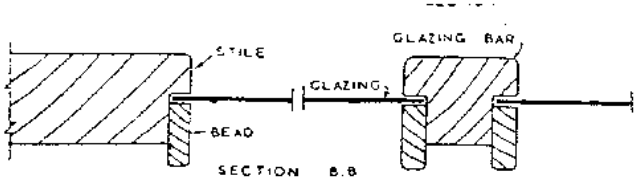
Partly Glazed Door



ELEVATION



VERTICAL
SECTION



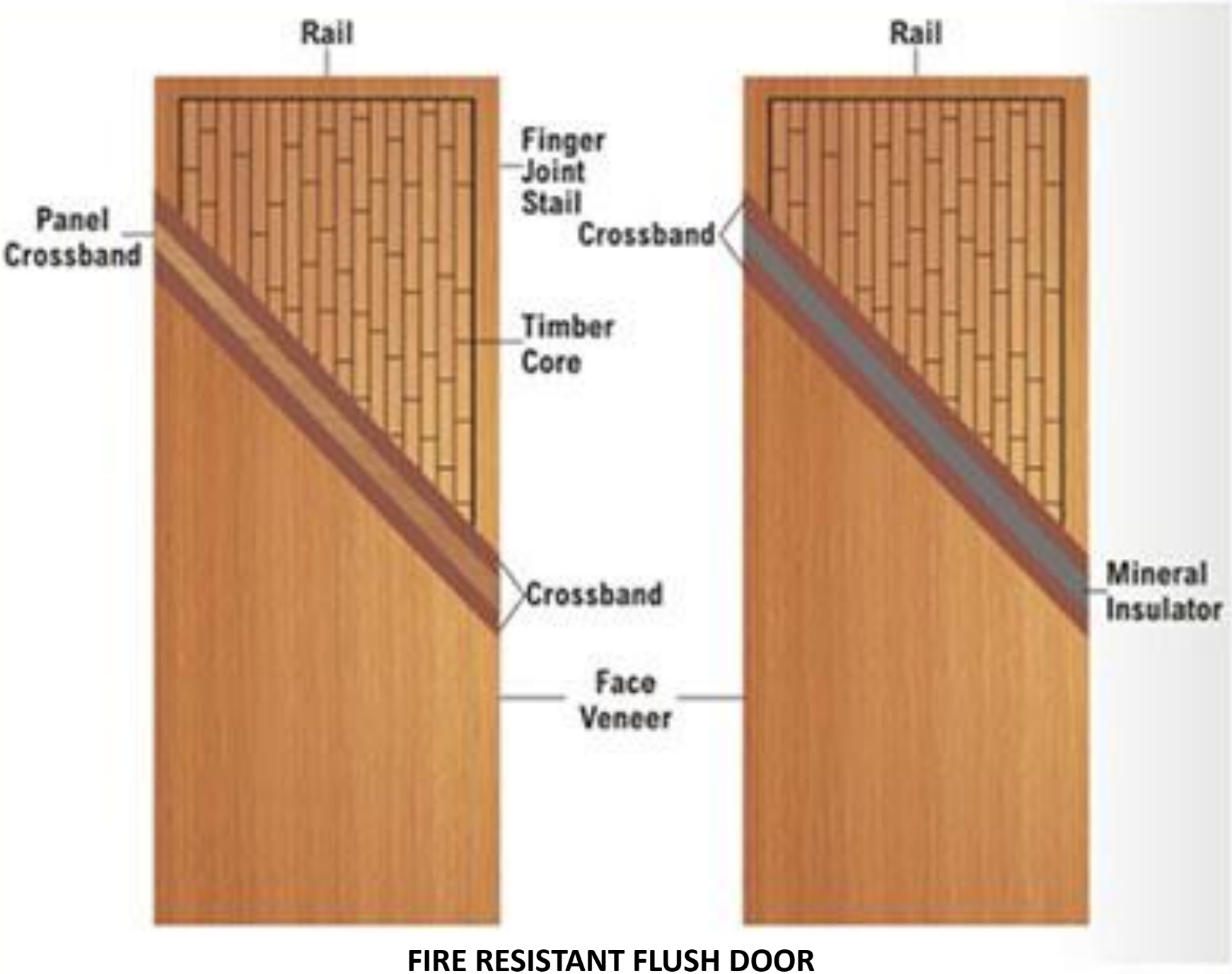
PART GLYLAZED AND PARTLY PANELED DOOR WITH GUN-STOCK RAIL

Flush Door

- Flush doors are simply doors with a completely flat surface on both sides.
- Flush doors can come in solid format which is a door made of solid wood or hollow format which is lightweight and comprised of two layers of thin timber separated, usually, by a lightweight honeycomb core. The core is covered with either hardboard or plywood on both sides.
- Solid flush doors are usually used as fire-check doors.
- Flush doors are lighter and cheaper than other types.

The flush door shutters are manufactured in standard thickness of 25, 30, 35 and 40 mm.





FIRE RESISTANT FLUSH DOOR

Types:

- Solid core flush door
- Hollow core
- Cellular core

Solid core door consist of the framework of vertical stiles and top rail and bottom rail.

The core consists of wooden strips (width not more than 25 mm) glued together under high pressure OR block board or particle board.

Plywood sheets /cross bands/face veneers are glued under high pressure to the assembly of core on both faces.

Such doors are quite strong but heavy and require more material. Solid-core doors provide better sound insulation and have less tendency to warp.

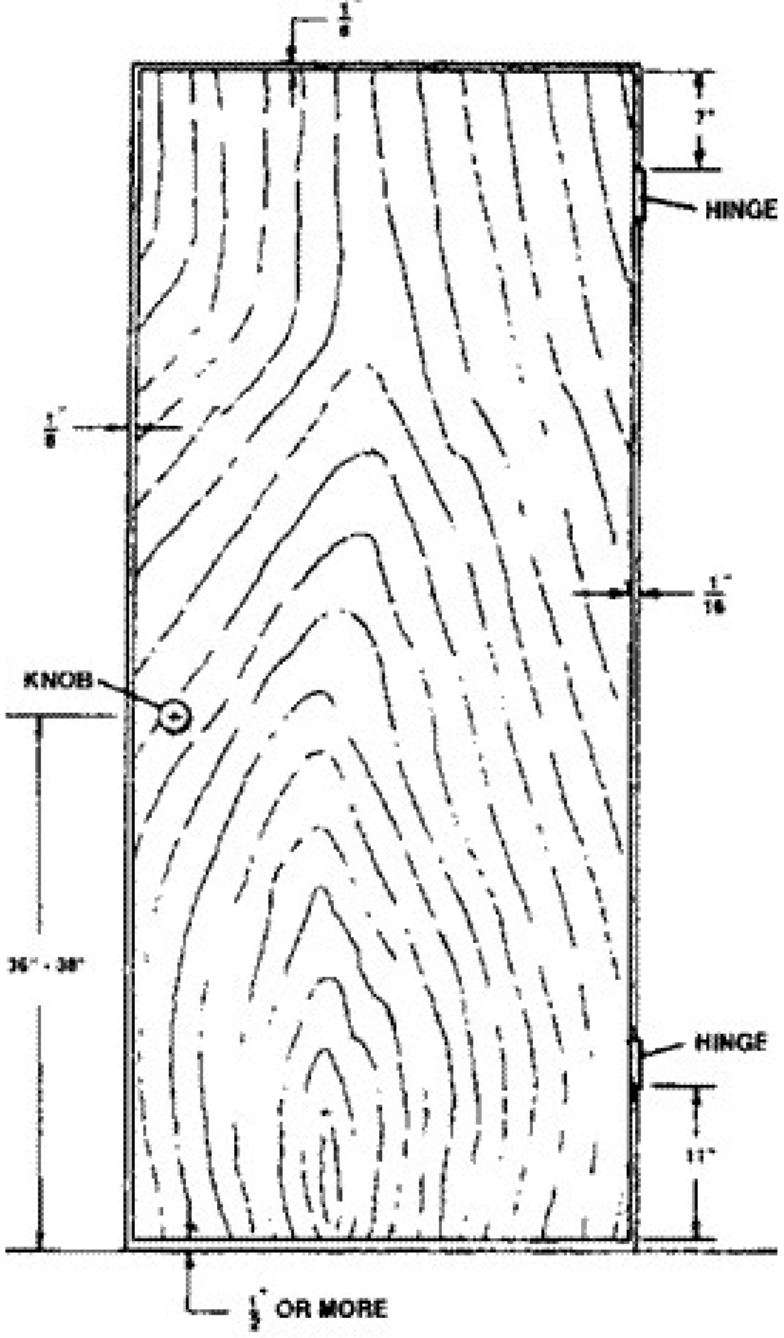


Figure 6-27.—Door clearances.

Hollow core

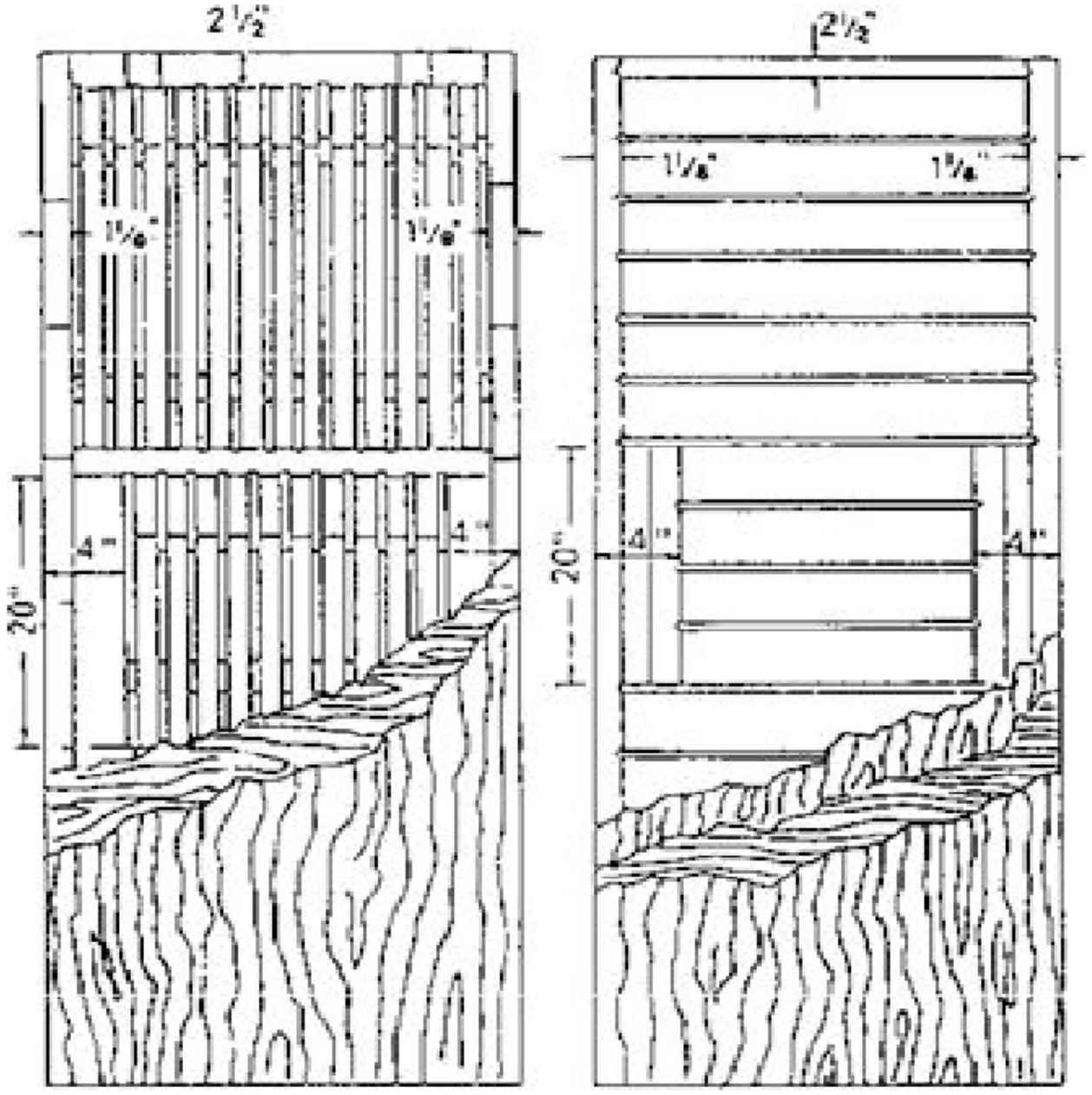
The frame consists of styles, top rail, bottom rail and two intermediate rails, not less than 75 mm in width.

The space between the styles and rails is filled by equally spaced battens each of minimum width 25 mm, such that voids are equally distributed and limited to 0.5 sq. m.

Plywood/veneers and cross bands can be glued from both sides. The thickness of plywood should not be less than 6 mm.

Cellular core

- The frame is made up of styles and top and bottom rail.
- The space is filled with closely spaced battens of minimum 25 mm width such that the void area does not exceed 0.25 sq. m.
- Thus, the total area of voids should not exceed 40% of the area of the shutter.

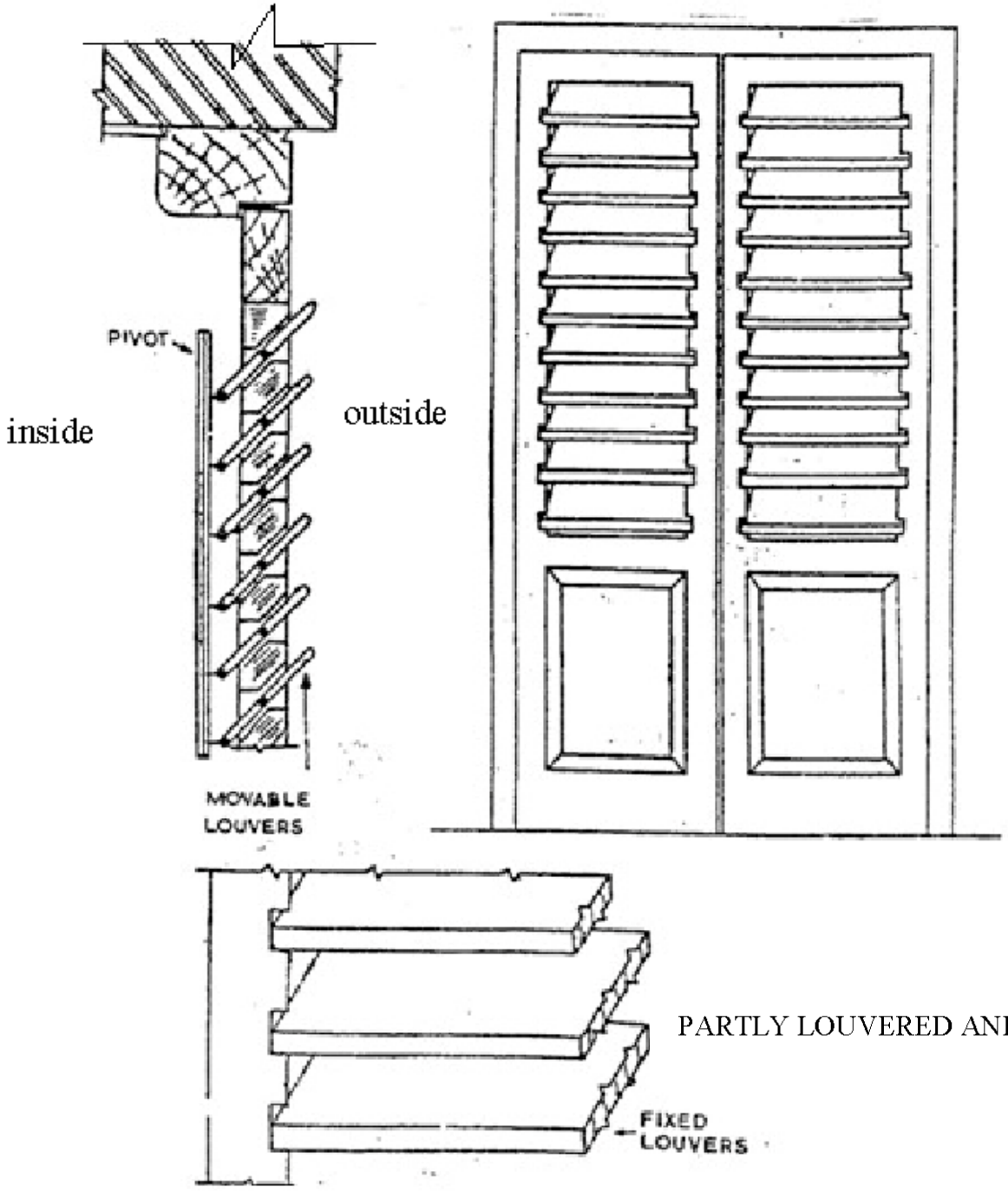


FIVE-PLY HOT PRESS HOLLOW CORE

SEVEN-PLY HOLLOW CORE

Louvered doors

- A **louvered door** has fixed or movable wooden louvers which permit open ventilation while preserving privacy and preventing the passage of light to the interior.
- They are most commonly used for bath and W.C. In residential and public buildings, where good ventilation is desired.
- The door may be louvered to its full height or may be partly louvered and partly paneled.
- The louvers are arranged in inclined fashion thus obstructs the vision but permits entry of air.
- Louvers may be fixed or movable.
- Louvers may be of timber, plywood or glass.
- However, they are difficult to clean.



PARTLY LOUVERED AND PARTLY PANELED DOOR





Typical
1 Louver
Elevation



Typical
2 Louver
Elevation



Typical
1 Louver over 1 Panel
Elevation

Revolving doors

- Such types are provided in public buildings, like banks, museums, hotels, offices etc.

  A **revolving door** normally has four wings/leaves that hang on a center shaft and rotate one way about a vertical axis within a round enclosure. The central shaft is fitted with ball bearing arrangement at the bottom, which allows the shutters to move without any jerk and making noise.

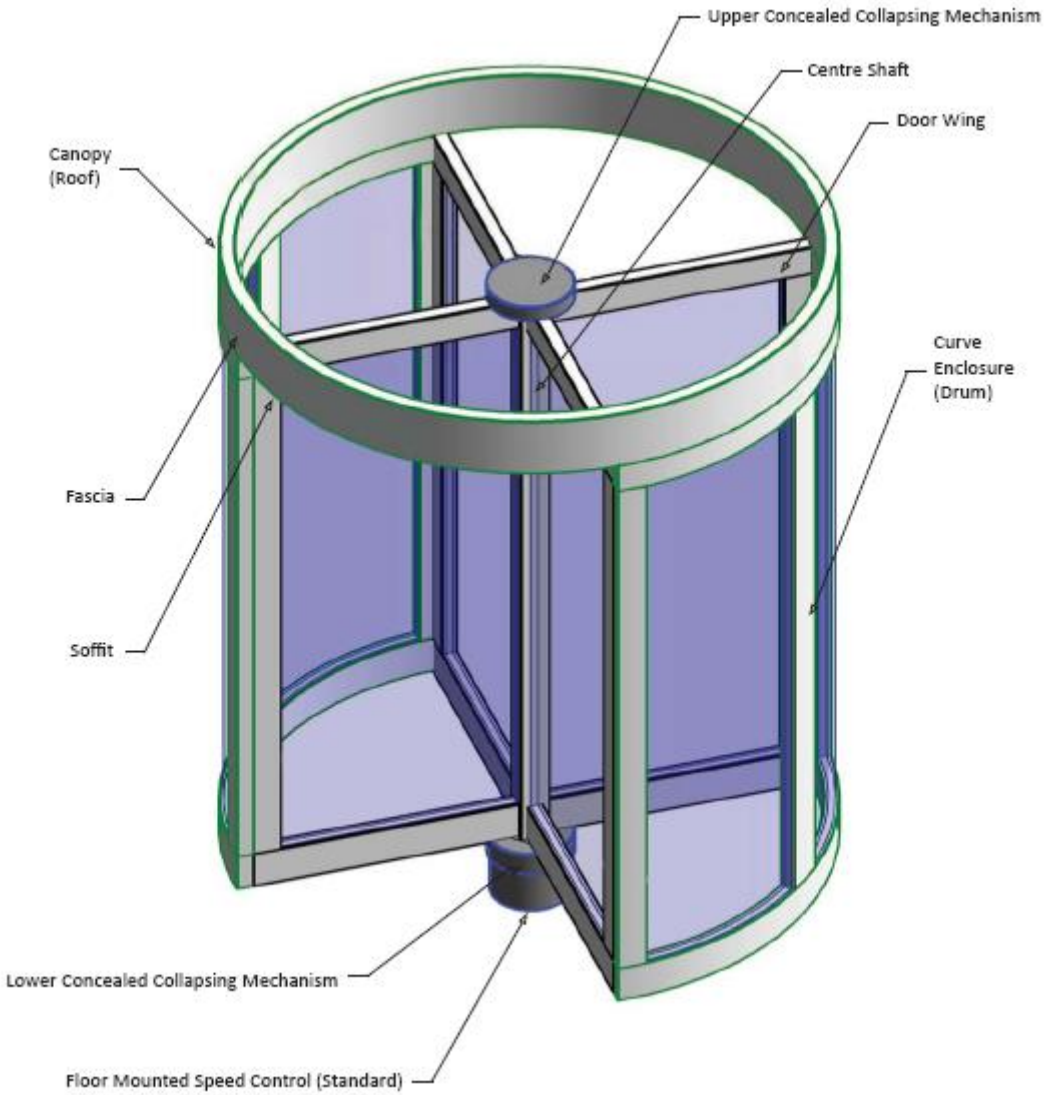
  The radiating shutters may be fully paneled, fully glazed or partly glazed. The glass doors allow people to see and anticipate each other while walking through. Vertical rubber pieces are provided at the rubbing end of the shutter to prevent drought of air.

People can walk out of and into the building at the same time.

The door closes automatically when not in use.

- Revolving doors are energy efficient by eliminating drafts, thus reducing the heating or cooling required for the building.
- The door may be motorized, or pushed manually using push bars. Revolving doors therefore create a good seal from the outside and help to reduce A/C and heating costs for climate control from the building.
- Revolving doors typically have a "speed control" to prevent people from spinning the doors too fast.




Upper Concealed Collapsing Mechanism
Centre Shaft
Door Wing
Canopy
(Roof)
Curve
Enclosure
(Drum)
Fascia
Soffit
Lower Concealed Collapsing Mechanism
Floor Mounted Speed Control (Standard)

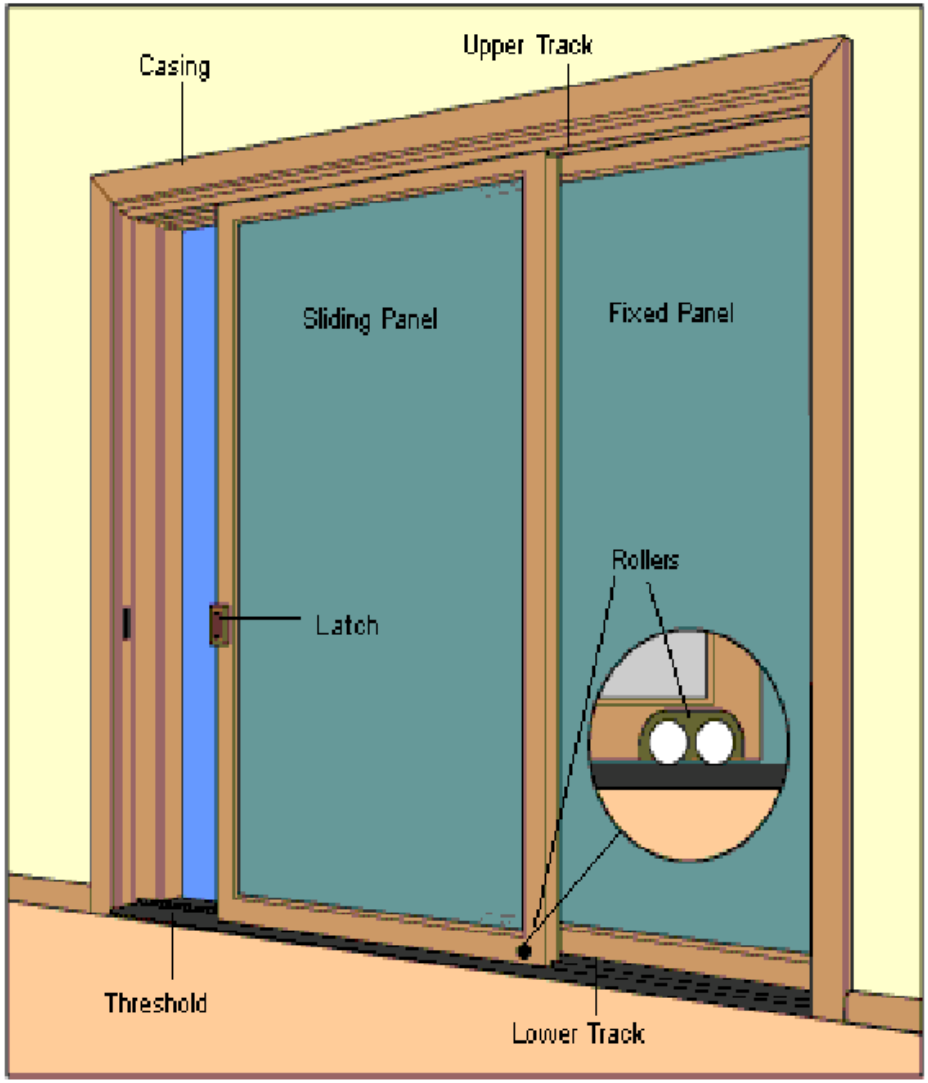
Sliding doors

- In these doors, the shutter slide horizontally along tracks with the help of runners and rails. often for space or
- Sliding glass doors are common in places where there is no space to swing the door.
- Such doors are very popular for use for the entrances to commercial structures and also in residential buildings for aesthetic considerations. .
- Sliding doors consist of either one, two or three doors that slide by each other on a track depending upon the size of opening and space available for sliding.
- They are pretty easily cleaned and maintained.
- These doors sound insulation is pretty poor usually, and they must be of high quality and fitted exactly in their tracks or else they may slide out of them.

- When fully open these doors will allow half the space of the opening in double sliding doors, or one third if triple.
- Sliding doors move along metal, wood, or vinyl tracks fitted into their frames at the top and bottom. To ease their movement, sliding doors often have plastic rollers attached to the top and bottom or to the bottom only.
- The door is hung by two trolley hangers at the top of the door running in a concealed track while at the bottom, rollers are provided to slide the shutter in a channel track.






Casing
Upper Track
Sliding Panel
Fixed Panel
Rollers
Latch
Threshold
Lower Track

Swing doors

- The shutter is fitted to its frame by special double action hinges.
- The hinges permits the shutter to move both ways, inward as well as outward.
- The doors are not rebated at the meeting styles.
- To open the door, a slight push is made and the spring action brings the shutter in closed position.

The return of the shutter is with force and thus, the door shall be either fully glazed Or provided with a peep hole at eye level, to avoid accidents.







Folded doors

Made of many narrow vertical strips or creases that fold back to back into a compact bundle when doors are pushed open, these strips or creases will be hanged from the top, and run on a track. They save space as they do not swing out of the door opening, though their sound and weather isolation is poor. Folding doors are usually pretty noisy, and considered not so durable








CONCH

Collapsible Door

- Such doors are used in garages, workshops, public buildings etc. to provide increased safety and protection to property.
- The doors do not require hinges to close or open the shutter nor the frame to hang them.
- It acts like a steel curtain.
- The door is made up from vertical double channels (20x10x2 mm), jointed together with the hollows on the inside to create a vertical gap.
- These channels are spaced at 100-120 mm apart and braced with diagonal iron flats.
- These diagonals allow the shutter to open or closed.
- The shutter operate between two rails, one fixed to the floor and other to the lintel.
- Rollers are mounted at the top and bottom.



Rolling shutter

- These are commonly used for shops, godowns, stores etc.
- The door shutter acts like a curtain and thus provides adequate protection and safety against fire and thefts.
- The shutter is made up of thin steel slabs called laths or slates about 1.25 mm thick interlocked to each other and coiled upon specially designed pipe shaft called drum mounted at the top.
- The shutter moves in two vertical steel guide channels installed at their ends.
- The channel is made up of steel sheets and deep enough to accommodate the shutter and to keep it in position.
- A horizontal shaft and spring in the drum which allow the shutter to coiled in or out.
- These may be manually operated for smaller openings (upto 10 sq.m.).
- Above 10 sq. m., they may be operated manually.



WINDOWS

Window details

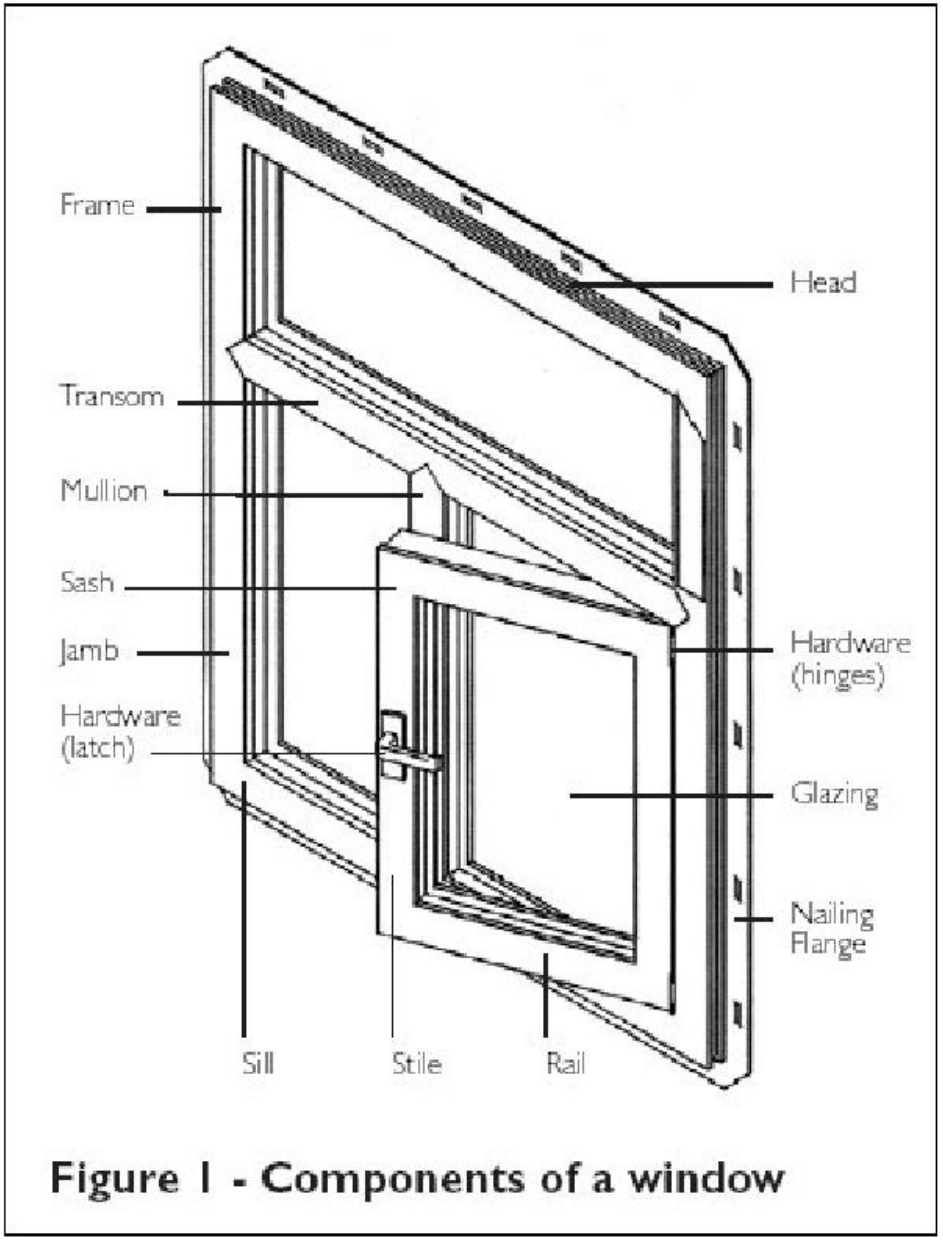
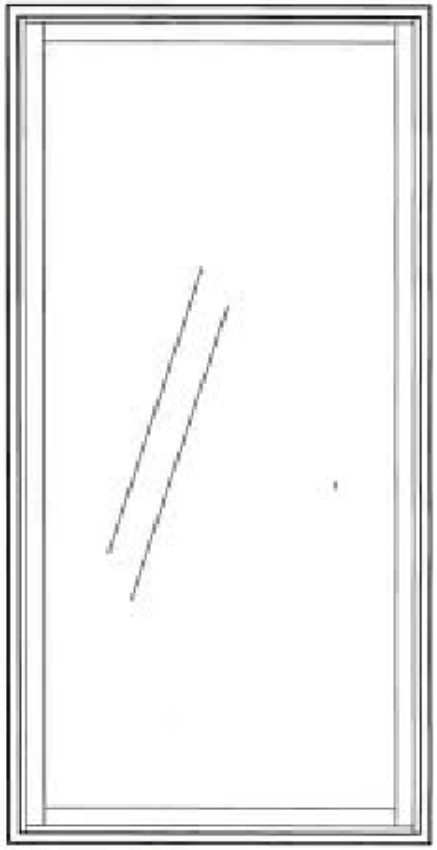


Figure 1 - Components of a window

TYPES OF WINDOWS

Fixed windows

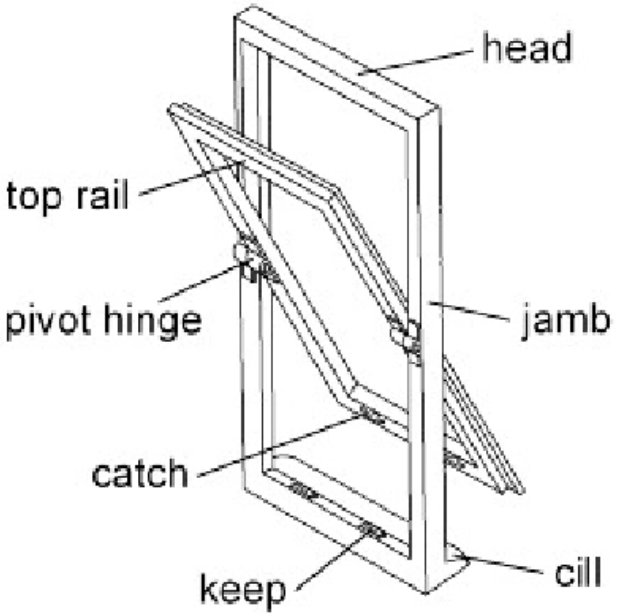
- In this type, the glass pane is permanently fixed in the opening of the wall.
- The shutter can't be opened or closed.
- The function is limited to allowing light and or permit vision in the room.
- No rebates are provided to the frame.
- The shutters are fully glazed.
- In homes they are generally decorative windows near doors, stairwells and high-places or are used in combination with other styles.





Pivoted windows

- In this type of window, the shutter is capable of rotating about a pivot fixed to window frame.
- The frame has no rebate.
- The shutter can swing horizontally or vertically.



pivot window

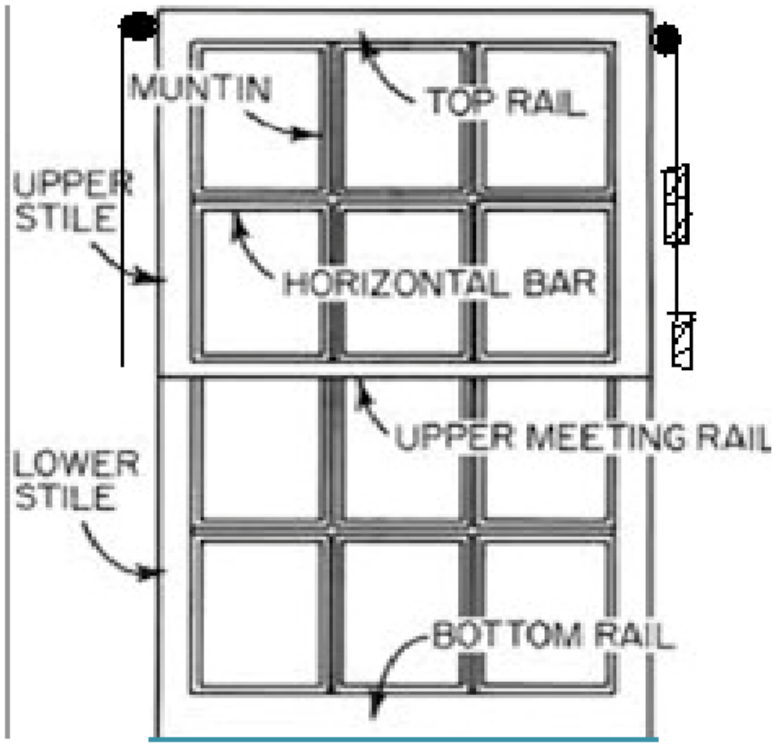


Horizontal pivoted

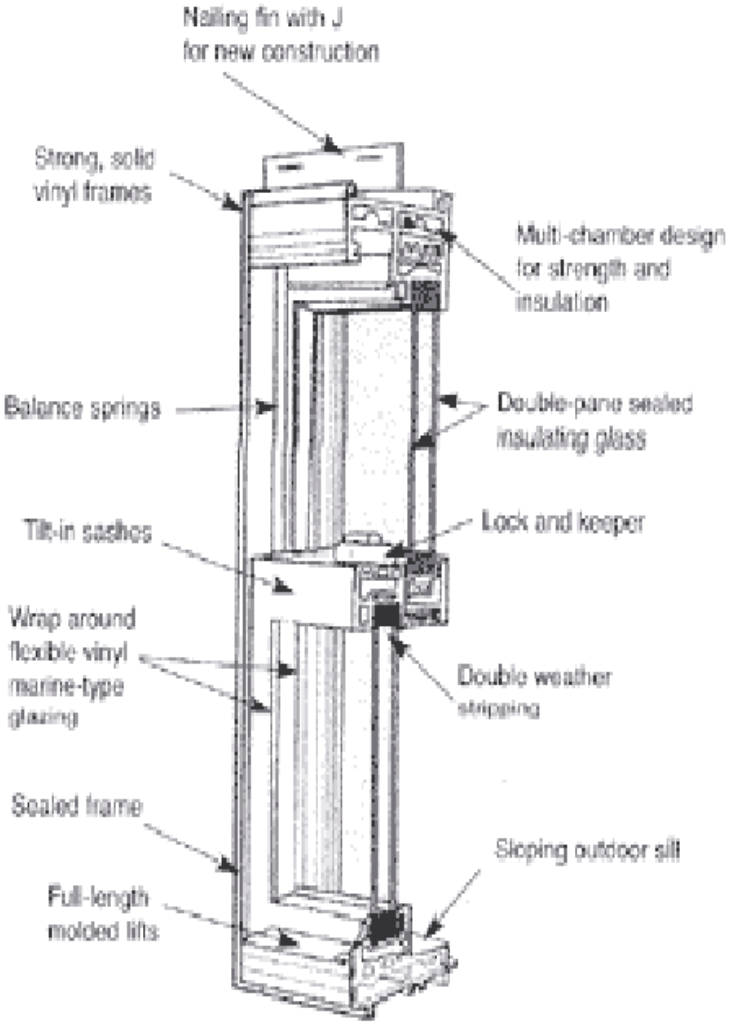
Vertical pivoted

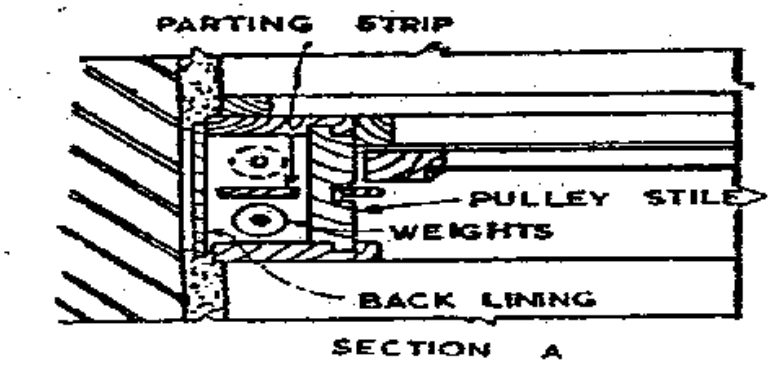
Double-hung windows

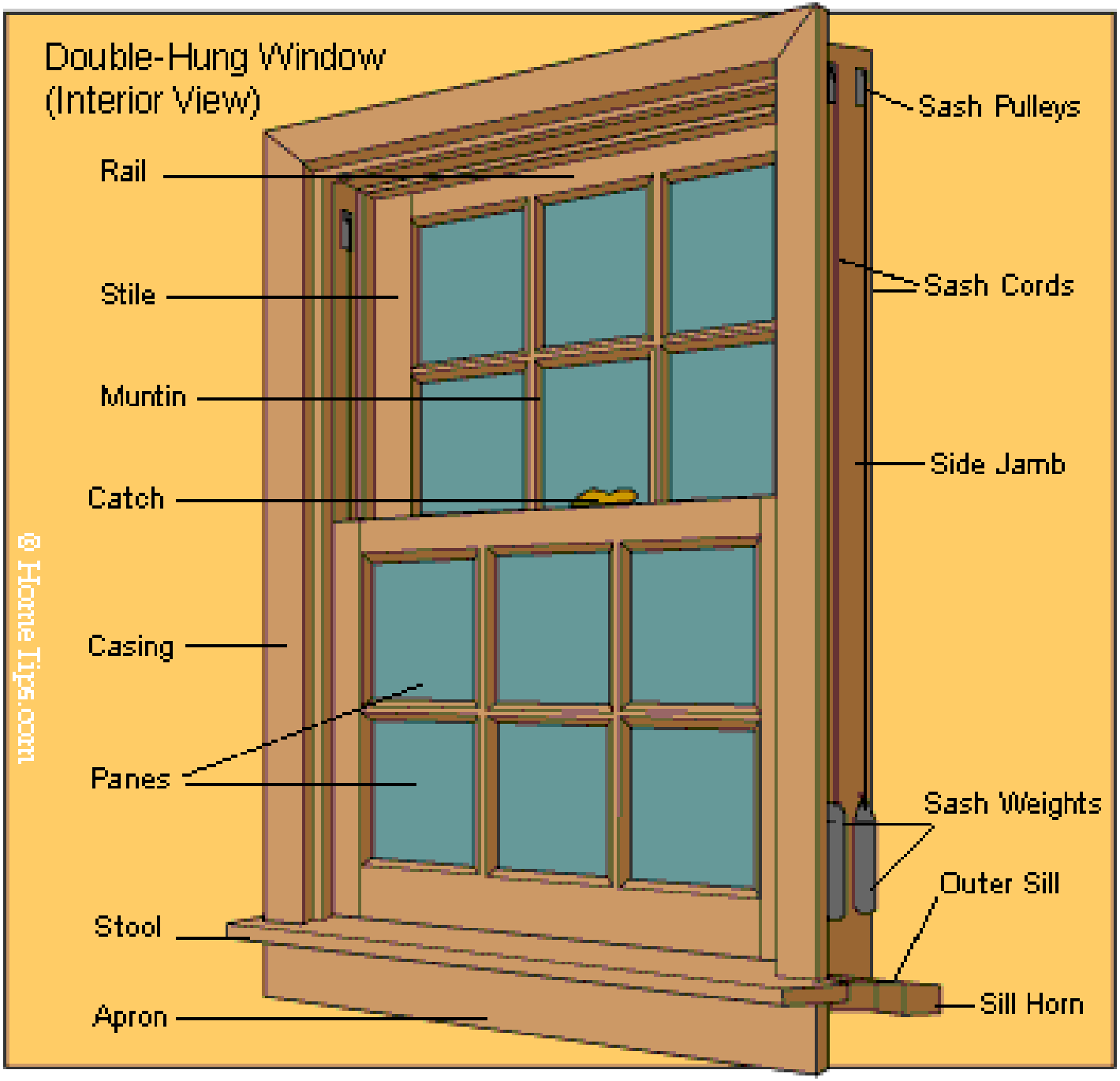
- It has two panes, top and bottom that slide up and down in tracks called stiles.
- The most common used windows today. When open, these windows allow air flow through half of its size.
- The two parts are not necessarily the same size.
- Traditionally, each shutter is provided with a pair of counterweights connected by cord or chain over pulleys.
- When the weights are pulled, the shutters open to required level.
- It is possible to have controlled ventilation.
- Sash windows may be fitted with simplex hinges which allow the window to be locked into hinges on one side, while the rope on the other side is detached, allowing the window to be opened for escape or cleaning.
- Nowadays, most new double-hung sash windows use spring balances to support the sashes.



COUNTER WEIGHT






Double-Hung Window
(Interior View)
Rail
Stile
Muntin
Catch
Casing
Panes
Stool
Apron
Sash Pulleys
Sash Cords
Side Jamb
Sash Weights
Outer Sill
Sill Horn
© Home Tips.com

Sliding Window or Slider:

- Has two or more sashes that overlap slightly but slide horizontally within the frame.
- Suitable openings or grooves are left in the frame or wall to accommodate the shutters when are shutters are opened.



Casement windows are hinged at the sides.

When fully opened, offer the maximum amount of ventilation.

Operates like a hinged door, except that it opens and closes with a lever inside the window.

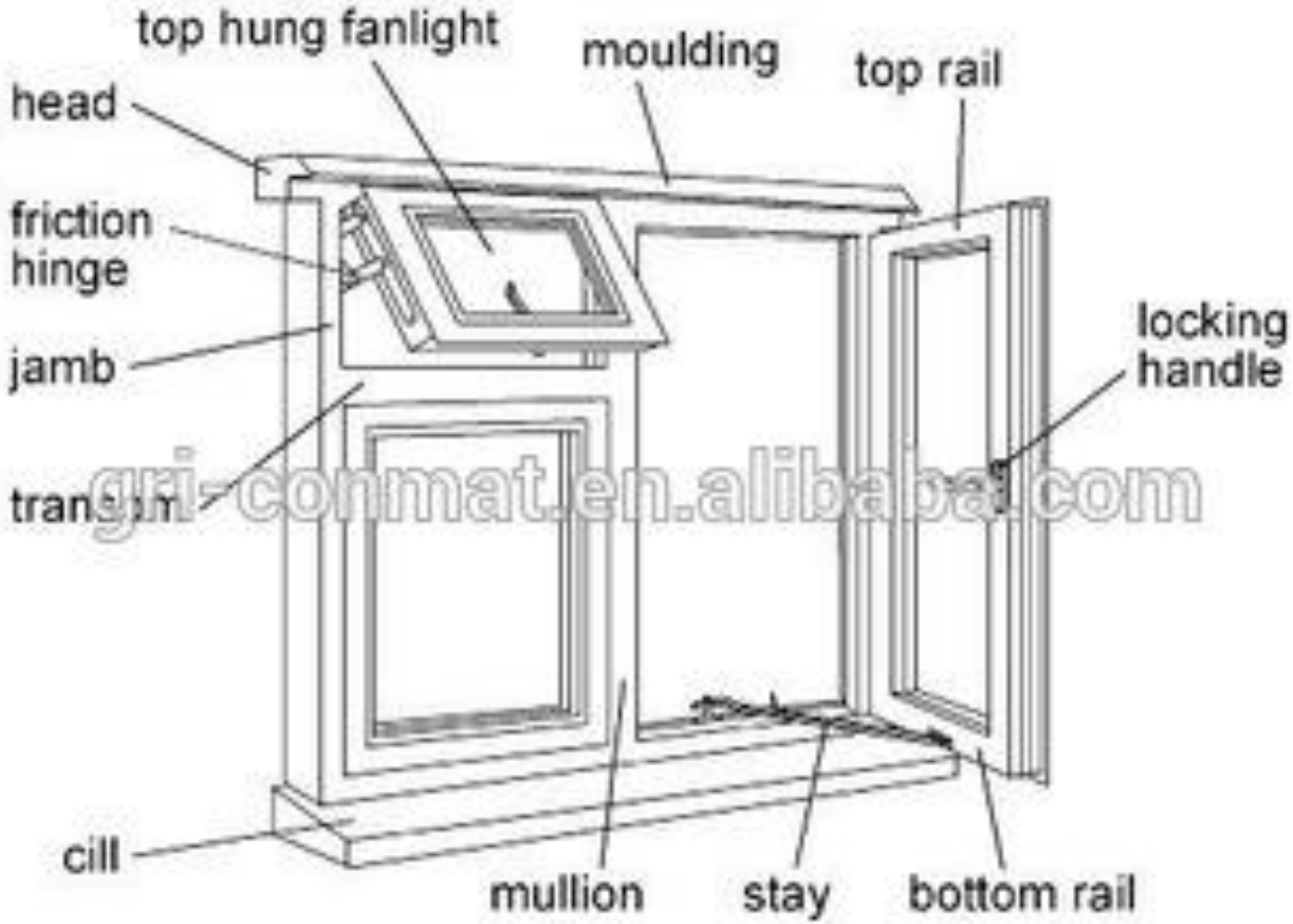
The shutter consists of styles, top rail, bottom rail and intermediate rail.

Depending upon the design, the frame can have additional vertical and horizontal members i.e. mullion and transom respectively.

The panels may be either glazed, unglazed or partly glazed and are fixed in the grooves made in rails and styles.







casement window

- Glazed window

This is a type of casement window where panels are fully glazed.

The frame has styles, top rail and a bottom rail.

The space between top and bottom rail is divided into number of panels with small timber members called, sash bars or glazing bars.

The glass panels are cut 1.5-3.0 mm smaller in size than the panel size to permit movement of sash bars.

Glass panes are fixed to sash bars by putty or by timber beads.









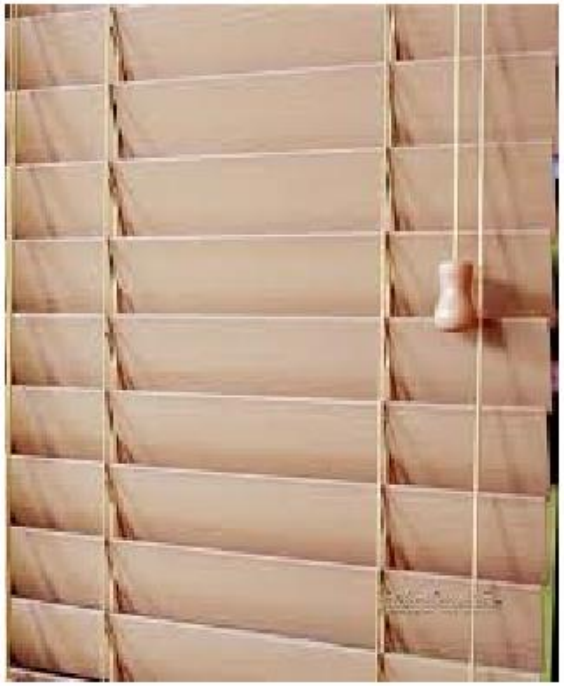
Louvered window

- They are provided for the sole function of ventilation and not for the vision outside.
- The styles are grooved to receive a series of louvers which may be of glass or wood slates.
- The louvers re usually fixed at 45^0 inclination sloping downward to the outside to run-off the rain water.
- The windows provide light and ventilation even if closed.
- Such windows are recommended for bath, WC, workshops etc., where privacy is more important.
- Venetian shutters uses louvers which can be opened or closed. The louvers are pivoted at both ends in the frame and in addition each blade is connected to a vertical batten by hinge.





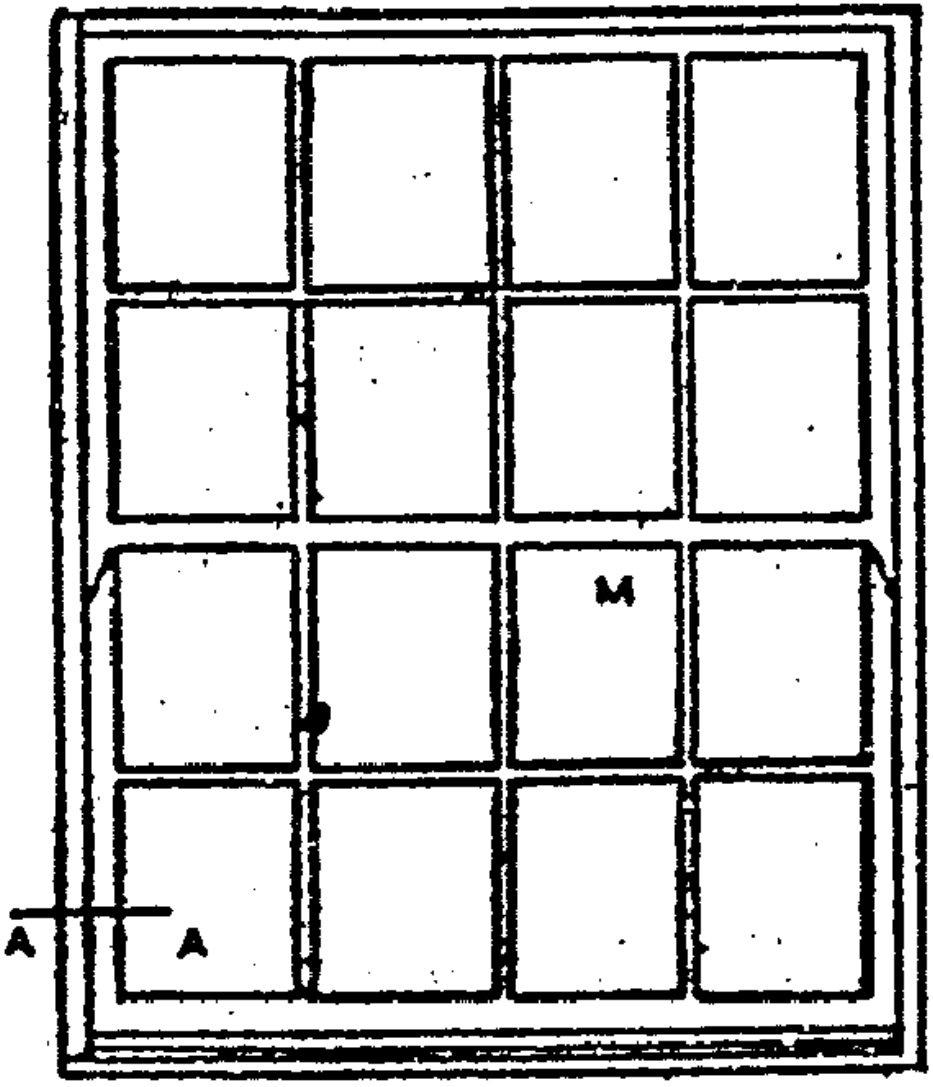




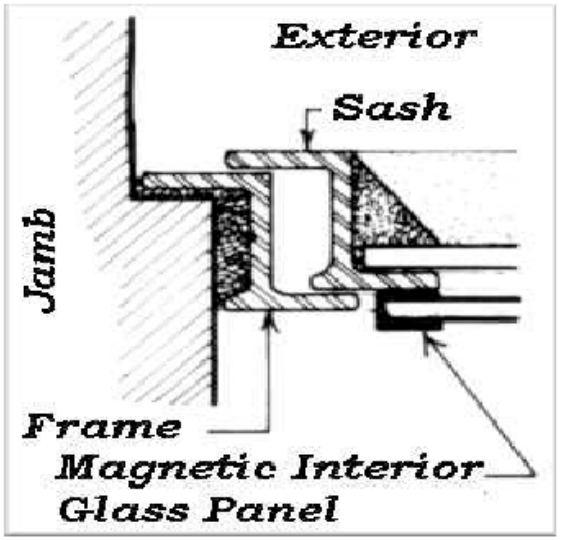


Metal Windows:

- These are very popular in public buildings and can be made up of mild steel, stainless steel, aluminium, bronze etc.
- Mils steel being cheapest of all, they are widely used. The windows can be fabricated for the required size using light rolled steel sections.
- They can be fixed directly to the wall opening in a wooden frame or in the steel frame.
- While fixing, care has to be taken that the members of the frame are not subjected to any structural loads to prevent damage.
- Thus, the size of the window opening is kept slightly more than the frame size so as to allow some clearance between the two.
- The window is fixed into the opening only after masonry and lintel work is over and fully set.


M
A
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ELEVATION


Exterior
Sash
Jamb
Frame
Magnetic Interior
Glass Panel







Bay window

- The window projecting outward from the external walls .
- Wide and decoratively impressive allow for 180° view.
- A multi-panel window, with at least three panels set at different angles to create an extension from the wall line.
- it is commonly used in cold country where snow often falls.
- They may be triangular, circular, rectangular or polygonal in plan.







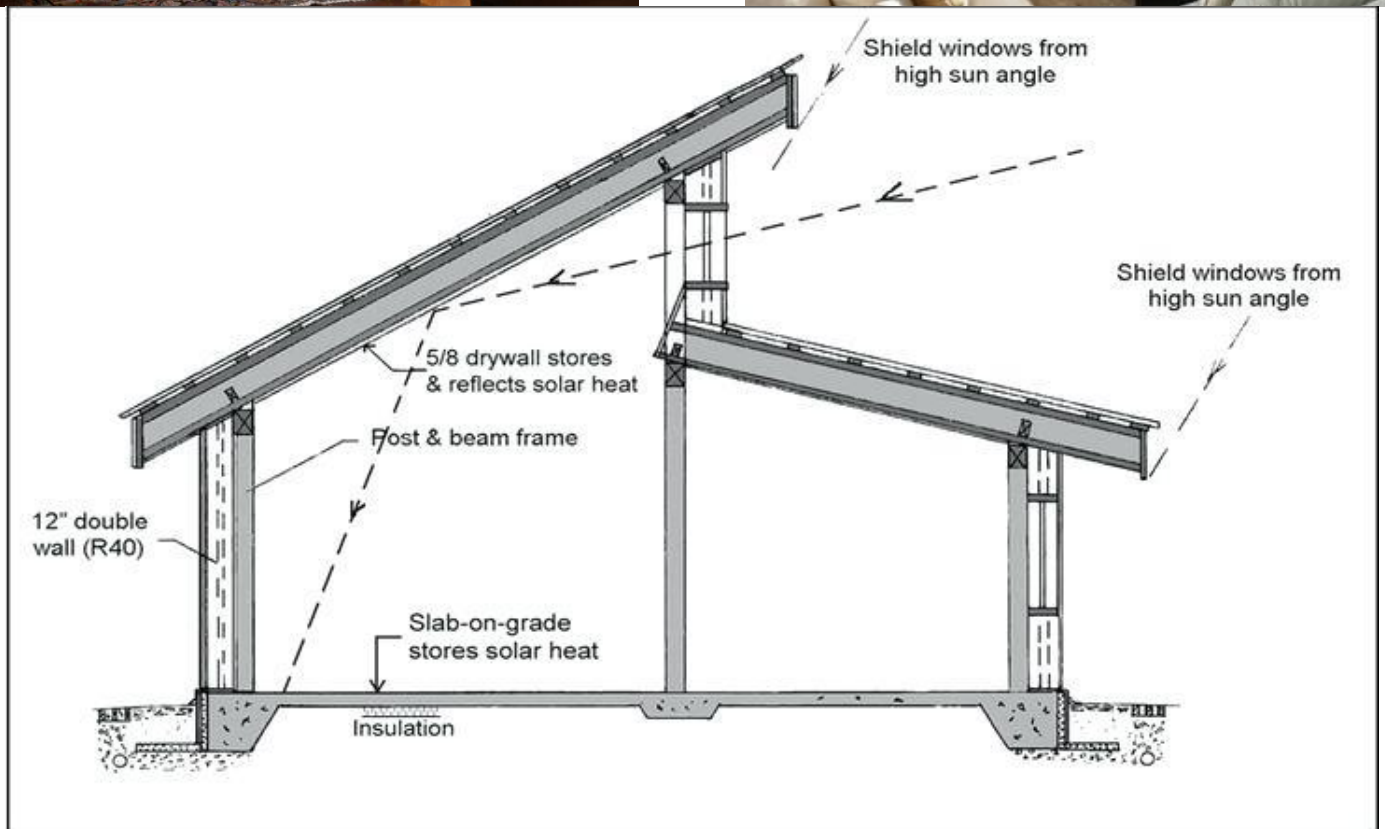


Clerestory window

- These are provided to permit light and ventilation to a room having more height than the adjoining rooms or when the ventilation is restricted.
- Generally provided near the top of main roof and they open above the slab of adjoining rooms.
- The shutters are generally pivoted at centre.
- The shutter can be opened or closed by means of two chords, each attached to the rails of the shutter.
- The shutter must swing in such a way that the upper part opens inside the room and lower part opens outside, to exclude rain water.






Shield windows from high sun angle
Shield windows from high sun angle
5/8 drywall stores & reflects solar heat
Post & beam frame
12" double wall (R40)
Slab-on-grade stores solar heat
Insulation

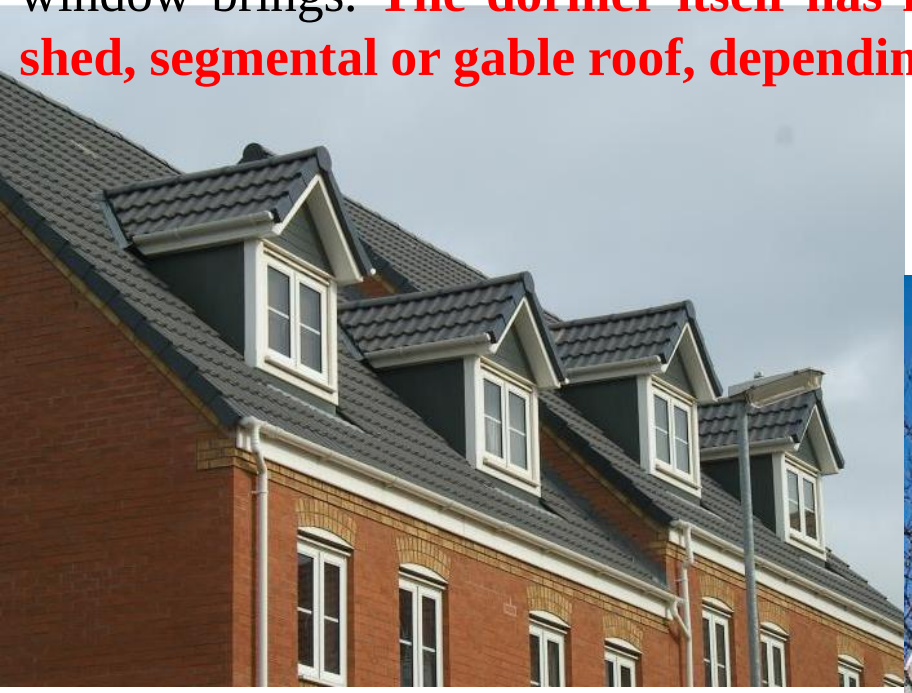
Corner window

- These are provided at the corner of the room.
- Light and air is admitted from two directions.
- The jamb post at the corner is made of heavy section.



DORMER WINDOW

The dormer protrusion on a sloped roof is determined by the size of the window and the designer's preference. Some people add blind or false dormers as an architectural feature in homes. But homes with an attic conversion that features an extra bedroom or an office welcome the addition of light and extra headroom that a dormer window brings. **The dormer itself has its own roof, which can be a hip, eyebrow, shed, segmental or gable roof, depending on the architectural design.**





GABLE WINDOW

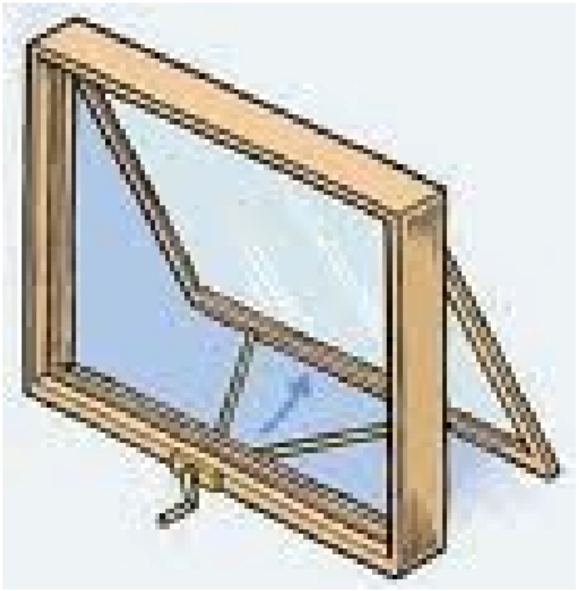
The gable window sits at the end of a pitched roof on the side of the house with a flat surface to allow natural light into the room. It doesn't require a protrusion because it is not placed into the sloping angle of the roof. It appears like a normal window set onto the house's side. The width of the house at the gable end determines how many windows can be placed in this area, as well as its proportions in relationship to the rest of the house



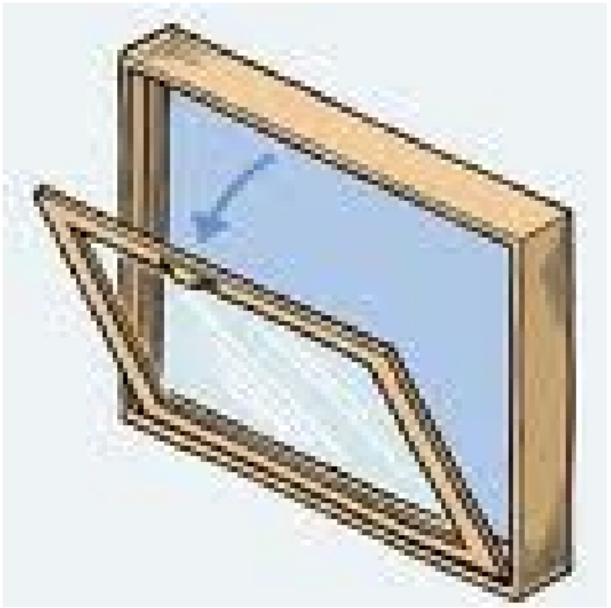


Awning windows are hinged at the top and open outward. They are designed to provide ventilation without letting in rain, etc.

- Awning windows can be used alone or in vertical or horizontal groups in combination with additional awning windows, other types of windows, or above doors.



Awning windows



Hopper windows




©Gardensheds, Inc.



Skylight

These are fixed windows on the sloping roofs.

- Admit natural light and help distribute light more evenly throughout the room. Considered an energy saver feature.
- In addition to reducing the need to use electric lights, it can deliver warmth in the winter and cooling in the summer, minimizing the need for fuel-based heating and air conditioning. On winter days, the sun's radiant energy can shine through a south- or west-facing skylight to warm interior surfaces. And in the summer, a ventilating skylight can promote air circulation by releasing the warm air that naturally rises.
- The opening for the window is made by cutting common rafters. The framework consist of trimming pieces, curb frames, bottom rail and top rail. The opening is treated with lead flashings to ensure water proofing.
- Skylights may be plastic or glass, fixed or operable, and made in any number of sizes and styles.




iStockphoto



- Fanlights: The small window or ventilator fitted above the door or window frame separated by transom. The function is to ensure cross ventilation in the room even if the door or windows are closed. They also assist in admitting natural light.

A **fanlight** is a **window**, often semicircular or semi-elliptical in shape, with glazing bars or tracery sets radiating out like an open fan. It is placed over another **window** or a doorway, and is sometimes hinged to a transom.



- Ventilator: It is a narrow window of small height fitted near the roof of a room for ventilation. The construction is similar to the fanlights. They are horizontally pivoted.





